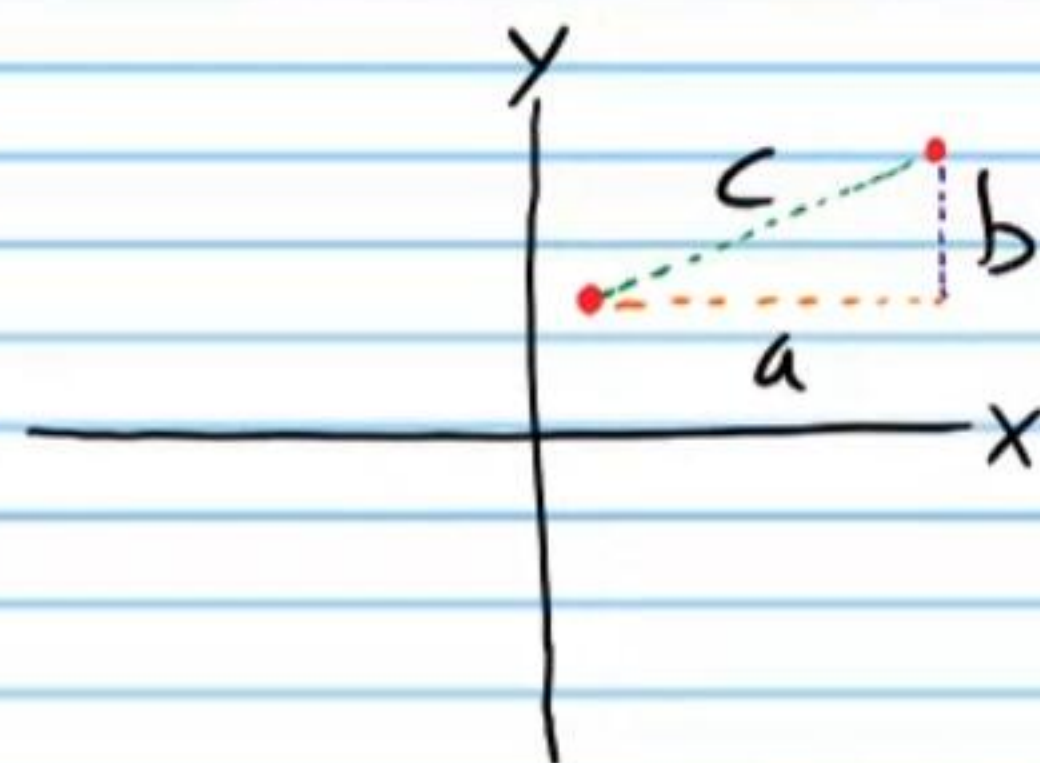
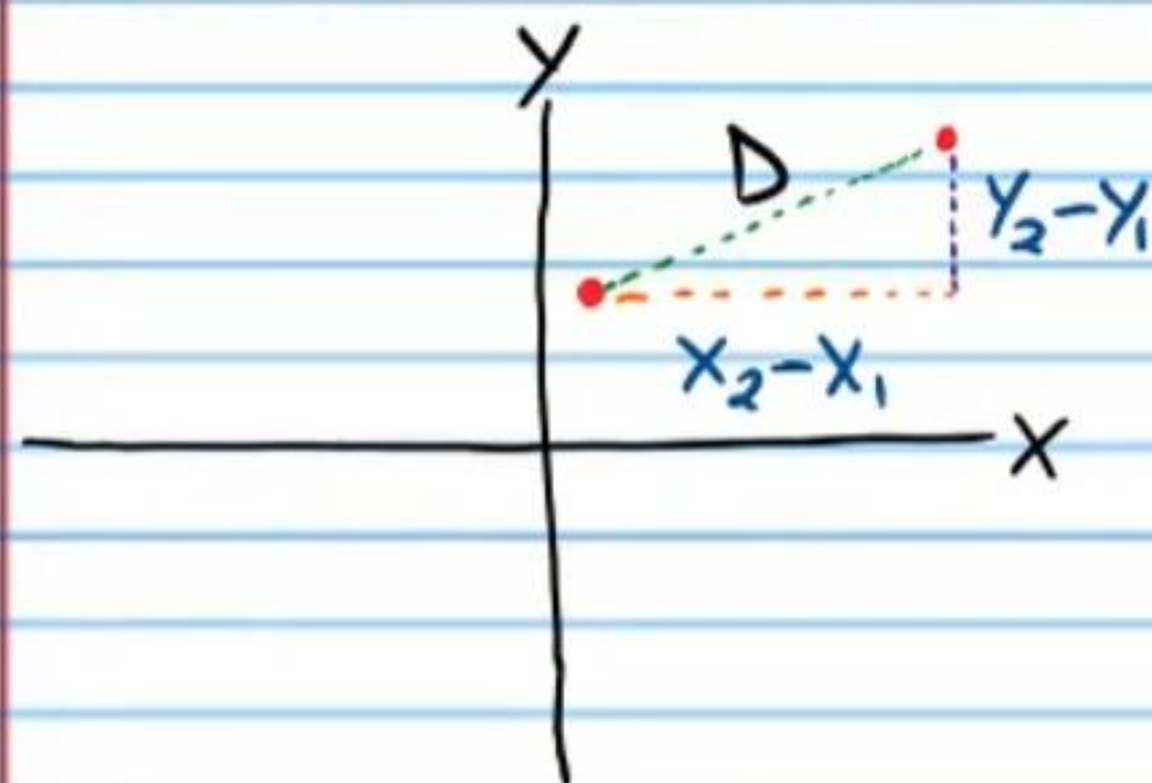
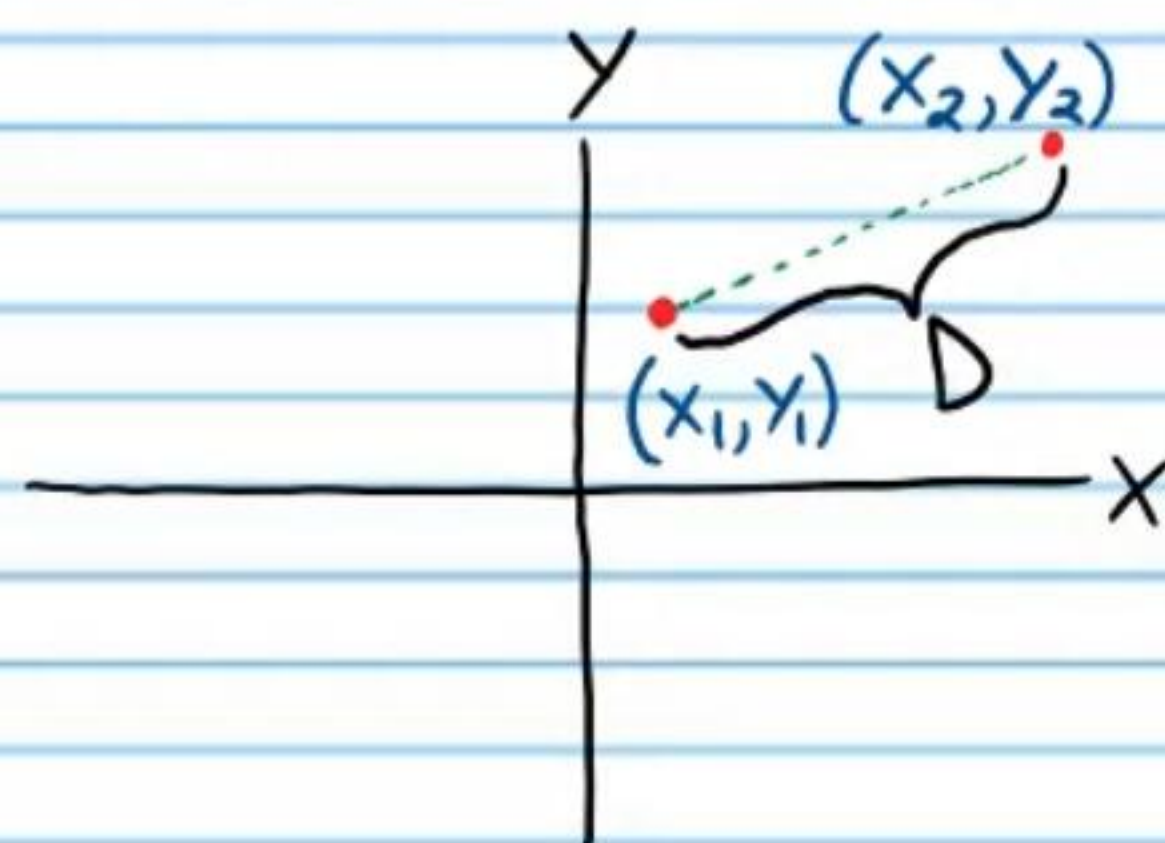


# The Distance Formula and the Midpoint in Two Dimensions

## The Distance Formula

The distance  $D$  between two points  $(x_1, y_1)$  and  $(x_2, y_2)$  is given by the following formula:

$$D = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$



$$c = \sqrt{a^2 + b^2}$$

$$c^2 = a^2 + b^2$$

Find the distance between the following points.

①  $(1, 3)$  &  $(7, 11)$

## Formula Method

$(x_1, y_1)$  &  $(x_2, y_2)$

$$D = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$D = \sqrt{(7 - 1)^2 + (11 - 3)^2}$$

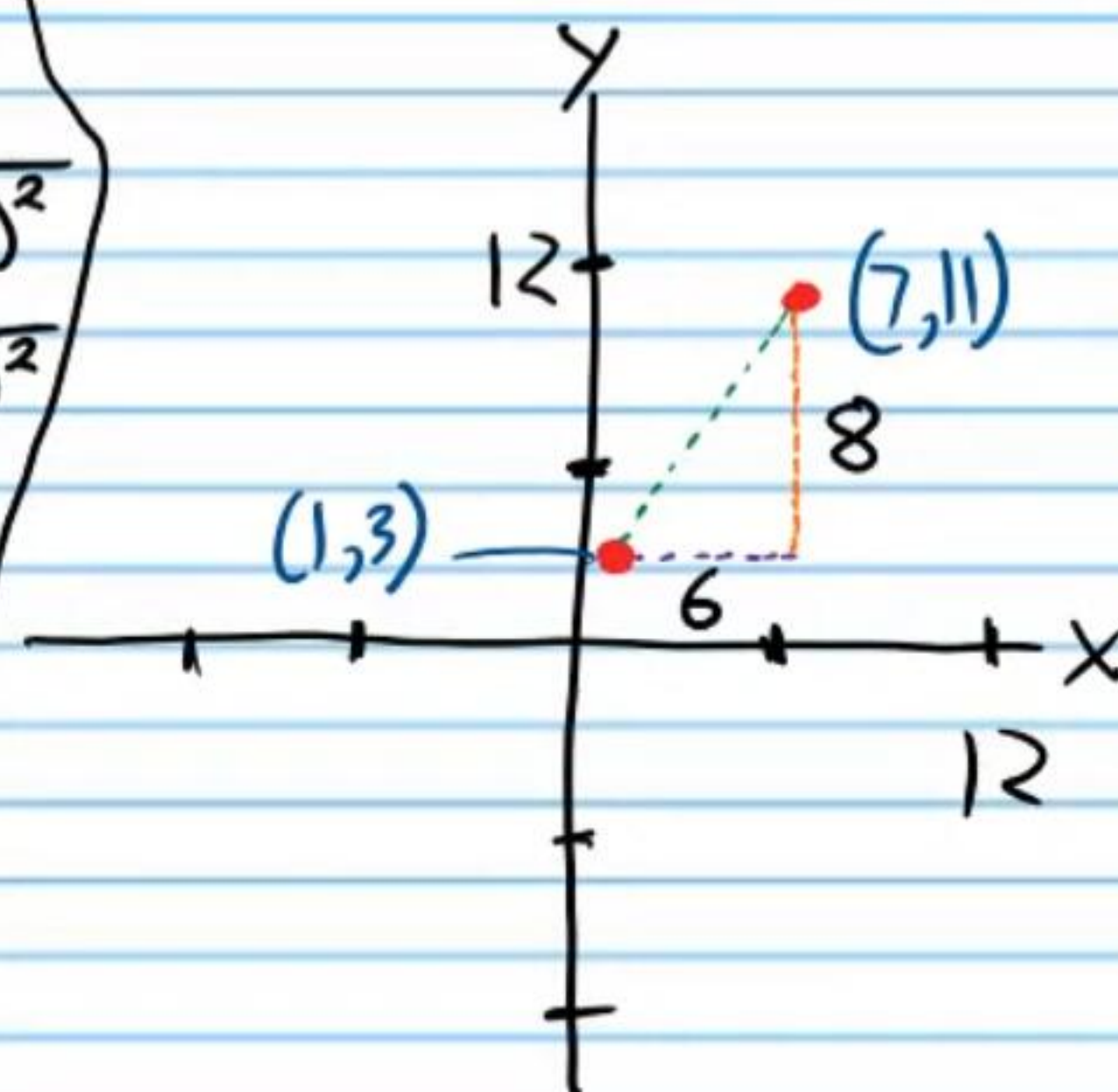
$$D = \sqrt{6^2 + 8^2}$$

$$D = \sqrt{36 + 64}$$

$$D = \sqrt{100}$$

$$\boxed{D = 10}$$

## Graphical Method



$$a^2 + b^2 = c^2$$

$$8^2 + 6^2 = c^2$$

$$64 + 36 = c^2$$

$$100 = c^2$$

$$\pm \sqrt{100} = \sqrt{c^2}$$

$$\pm 10 = c$$

$$10 = c$$

$$\boxed{10 = D}$$

②  $(-1, -2)$  &  $(-5, -7)$

Formula Method

$(X_1, Y_1)$  &  $(X_2, Y_2)$

$$D = \sqrt{(X_2 - X_1)^2 + (Y_2 - Y_1)^2}$$

$$D = \sqrt{(-5 - (-1))^2 + (-7 - (-2))^2}$$

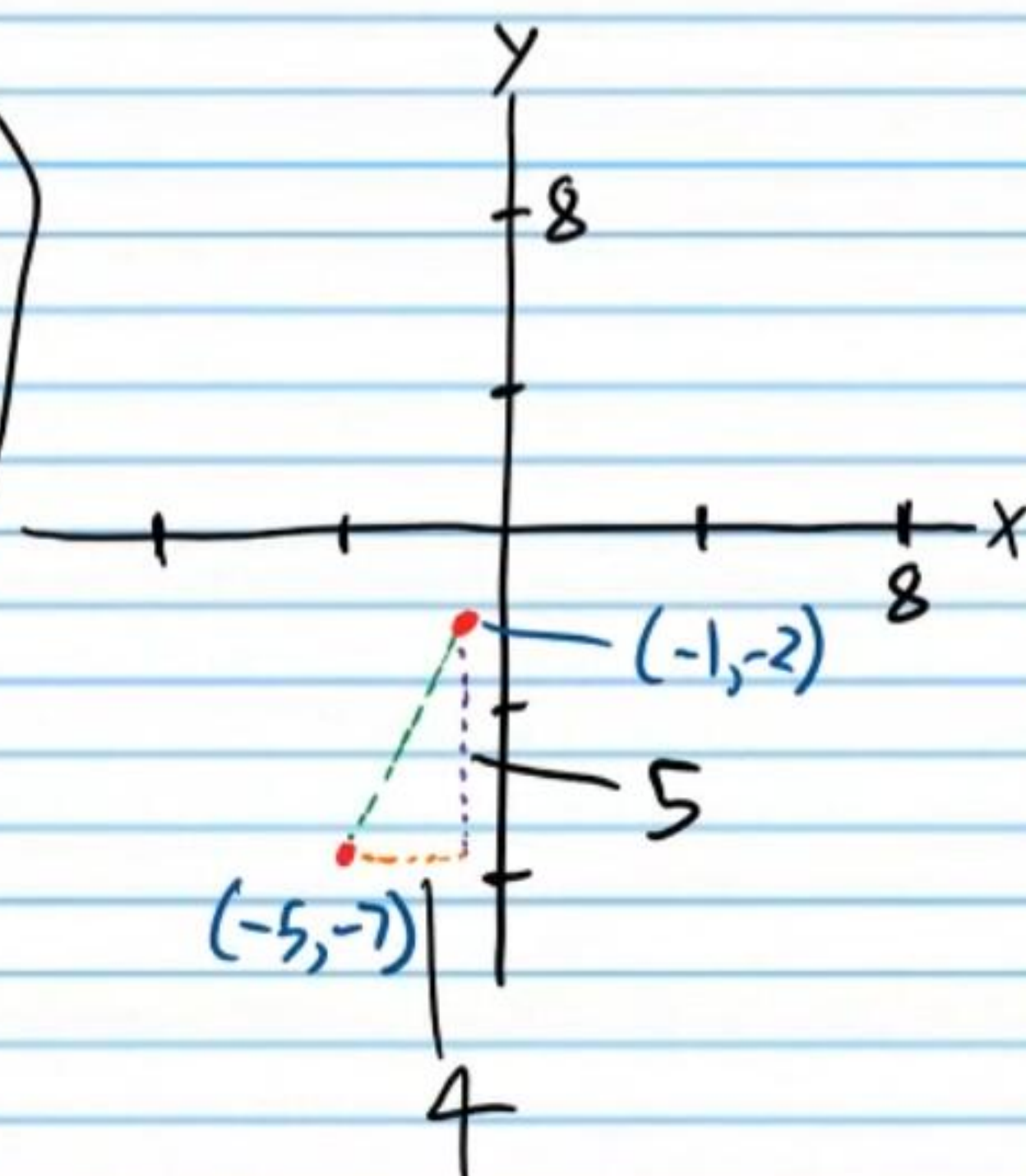
$$D = \sqrt{(-5 + 1)^2 + (-7 + 2)^2}$$

$$D = \sqrt{(-4)^2 + (-5)^2}$$

$$D = \sqrt{16 + 25}$$

$$\boxed{D = \sqrt{41}}$$

Graphical Method



$$a^2 + b^2 = c^2$$

$$5^2 + 4^2 = c^2$$

$$25 + 16 = c^2$$

$$41 = c^2$$

$$\sqrt{41} = \sqrt{c^2}$$

$$\sqrt{41} = c$$

$$\boxed{\sqrt{41} = D}$$

③  $(5, 1)$  &  $(-1, 3)$

Formula Method

$(X_1, Y_1)$  &  $(X_2, Y_2)$

$$D = \sqrt{(X_2 - X_1)^2 + (Y_2 - Y_1)^2}$$

$$D = \sqrt{(-1 - 5)^2 + (3 - 1)^2}$$

$$D = \sqrt{(-6)^2 + 2^2}$$

$$D = \sqrt{36 + 4}$$

$$D = \sqrt{40}$$

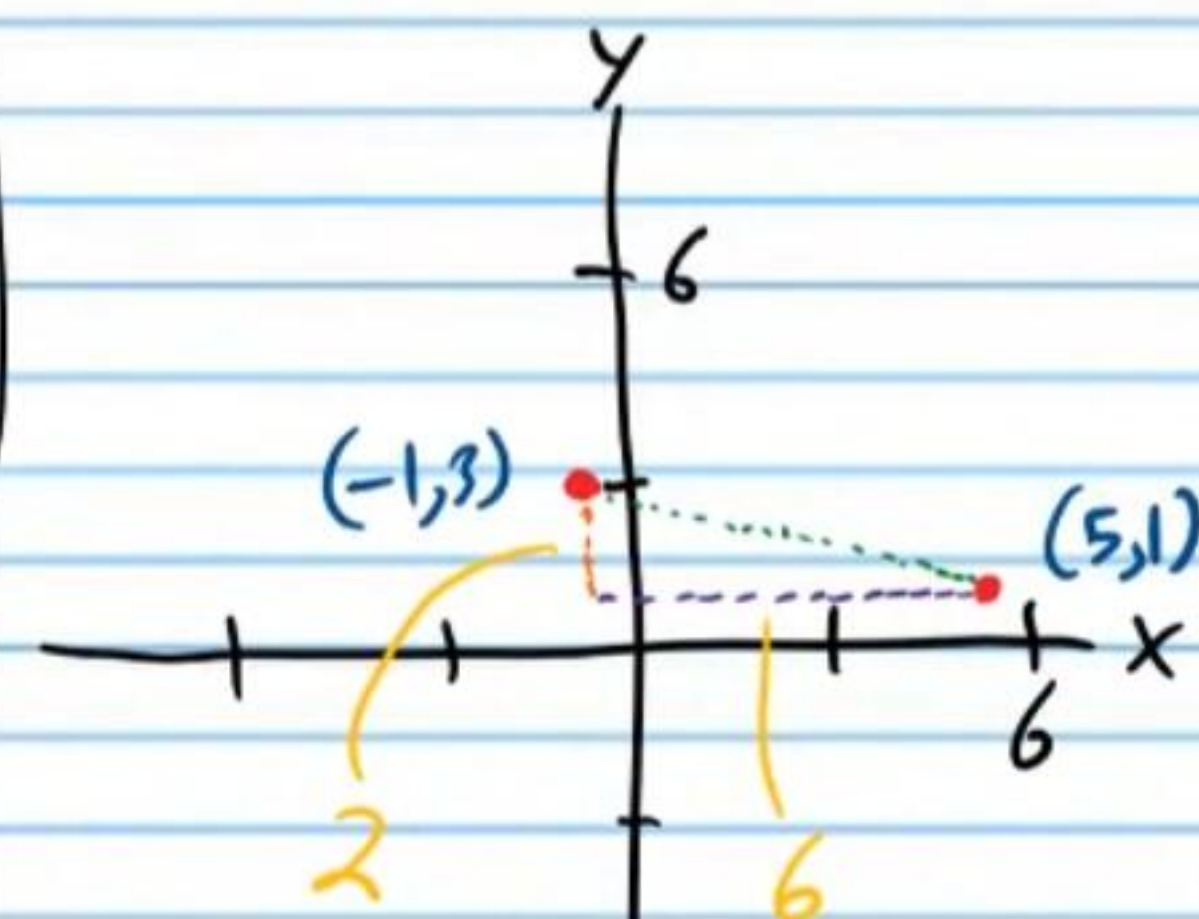
$$D = \sqrt{4 \cdot 10}$$

$$D = \sqrt{4} \cdot \sqrt{10} \quad \sqrt{a \cdot b} = \sqrt{a} \cdot \sqrt{b}$$

$$D = 2 \cdot \sqrt{10}$$

$$\boxed{D = 2\sqrt{10}}$$

Graphical Method



$$a^2 + b^2 = c^2$$

$$2^2 + 6^2 = c^2$$

$$4 + 36 = c^2$$

$$40 = c^2$$

$$\sqrt{40} = \sqrt{c^2}$$

$$\sqrt{40} = c$$

$$\sqrt{4 \cdot 10} = c$$

$$\sqrt{4} \cdot \sqrt{10} = c$$

$$2 \cdot \sqrt{10} = c$$

$$2\sqrt{10} = c$$

$$\boxed{2\sqrt{10} = D}$$

④ (5, -7) & (-4, 1)

Formula Method

$(x_1, y_1) \& (x_2, y_2)$

$$D = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$D = \sqrt{(-4 - 5)^2 + (1 - (-7))^2}$$

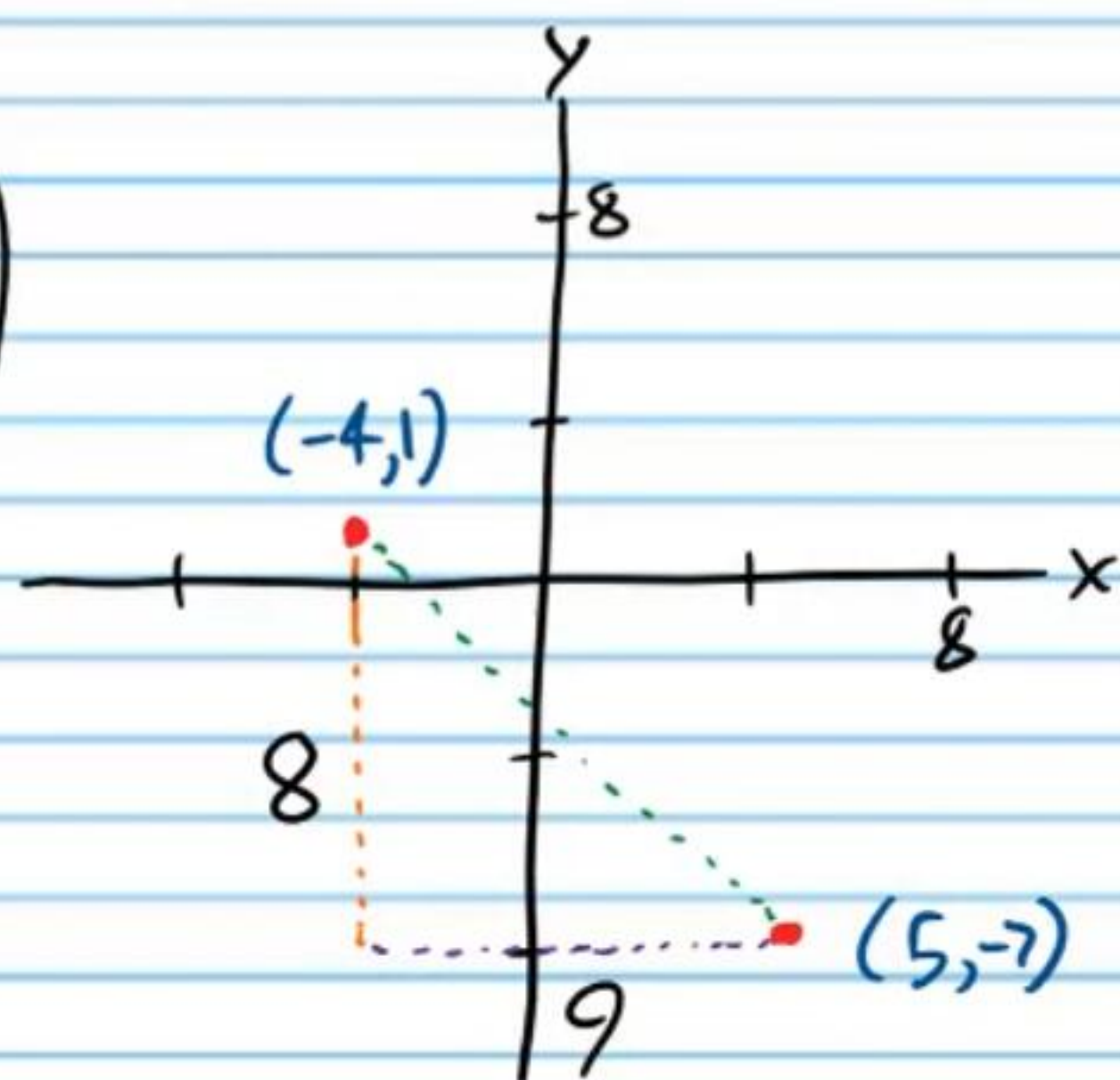
$$D = \sqrt{(-9)^2 + (1 + 7)^2}$$

$$D = \sqrt{81 + 8^2}$$

$$D = \sqrt{81 + 64}$$

$$\boxed{D = \sqrt{145}}$$

Graphical Method



$$a^2 + b^2 = c^2$$

$$8^2 + 9^2 = c^2$$

$$64 + 81 = c^2$$

$$145 = c^2$$

$$\sqrt{145} = \sqrt{c^2}$$

$$\sqrt{145} = c$$

$$\boxed{\sqrt{145} = D}$$

• ⑤ (3, 4) & (6, 8)

Formula Method

$(x_1, y_1) \& (x_2, y_2)$

$$D = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$D = \sqrt{(6 - 3)^2 + (8 - 4)^2}$$

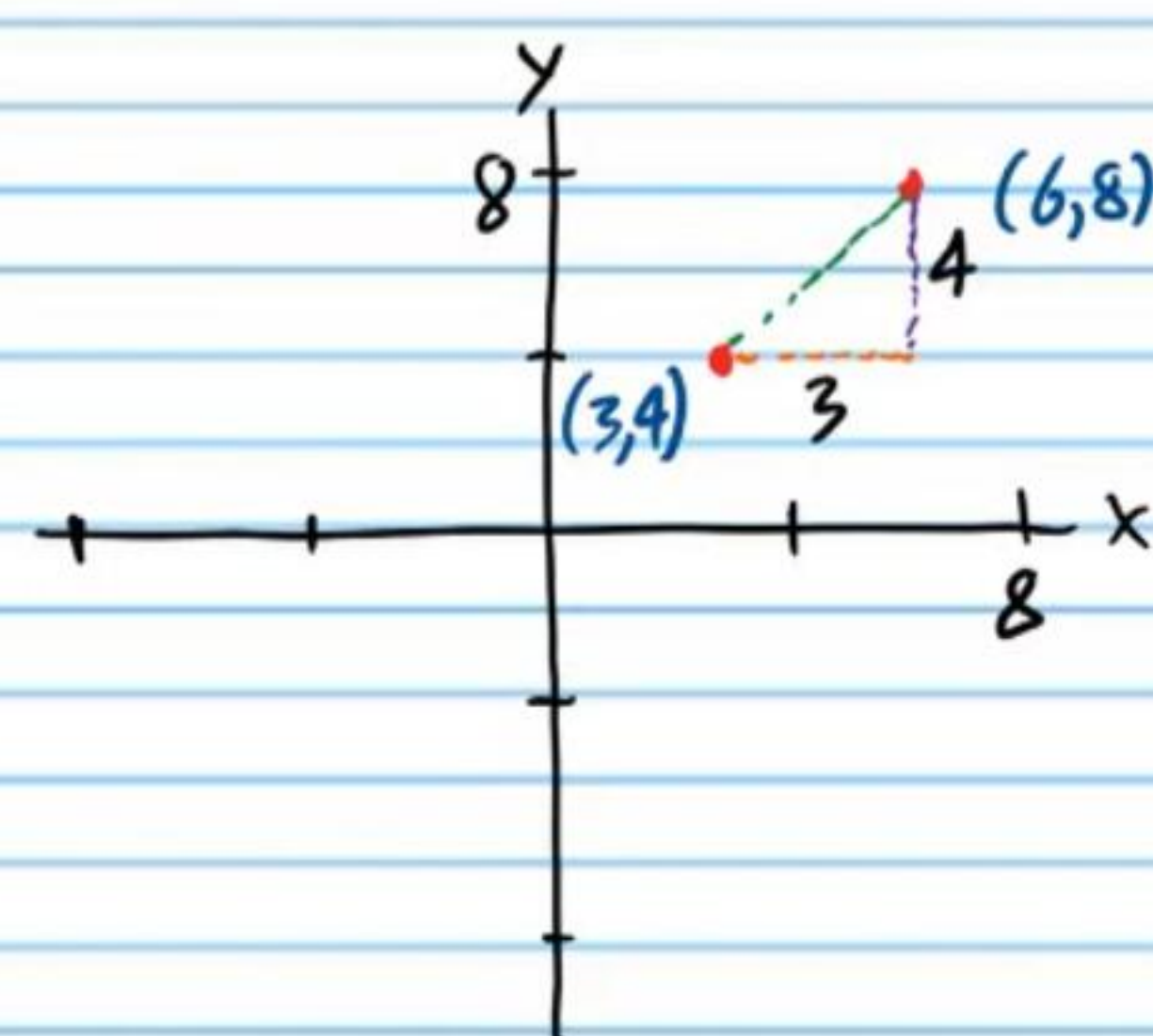
$$D = \sqrt{3^2 + 4^2}$$

$$D = \sqrt{9 + 16}$$

$$D = \sqrt{25}$$

$$\boxed{D = 5}$$

Graphical Method



$$a^2 + b^2 = c^2$$

$$3^2 + 4^2 = c^2$$

$$9 + 16 = c^2$$

$$25 = c^2$$

$$\sqrt{25} = \sqrt{c^2}$$

$$5 = c$$

$$\boxed{5 = D}$$

⑥  $(-6, 9)$  &  $(-8, 12)$

Formula Method

$(x_1, y_1)$  &  $(x_2, y_2)$

$$D = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$D = \sqrt{(-8 - (-6))^2 + (12 - 9)^2}$$

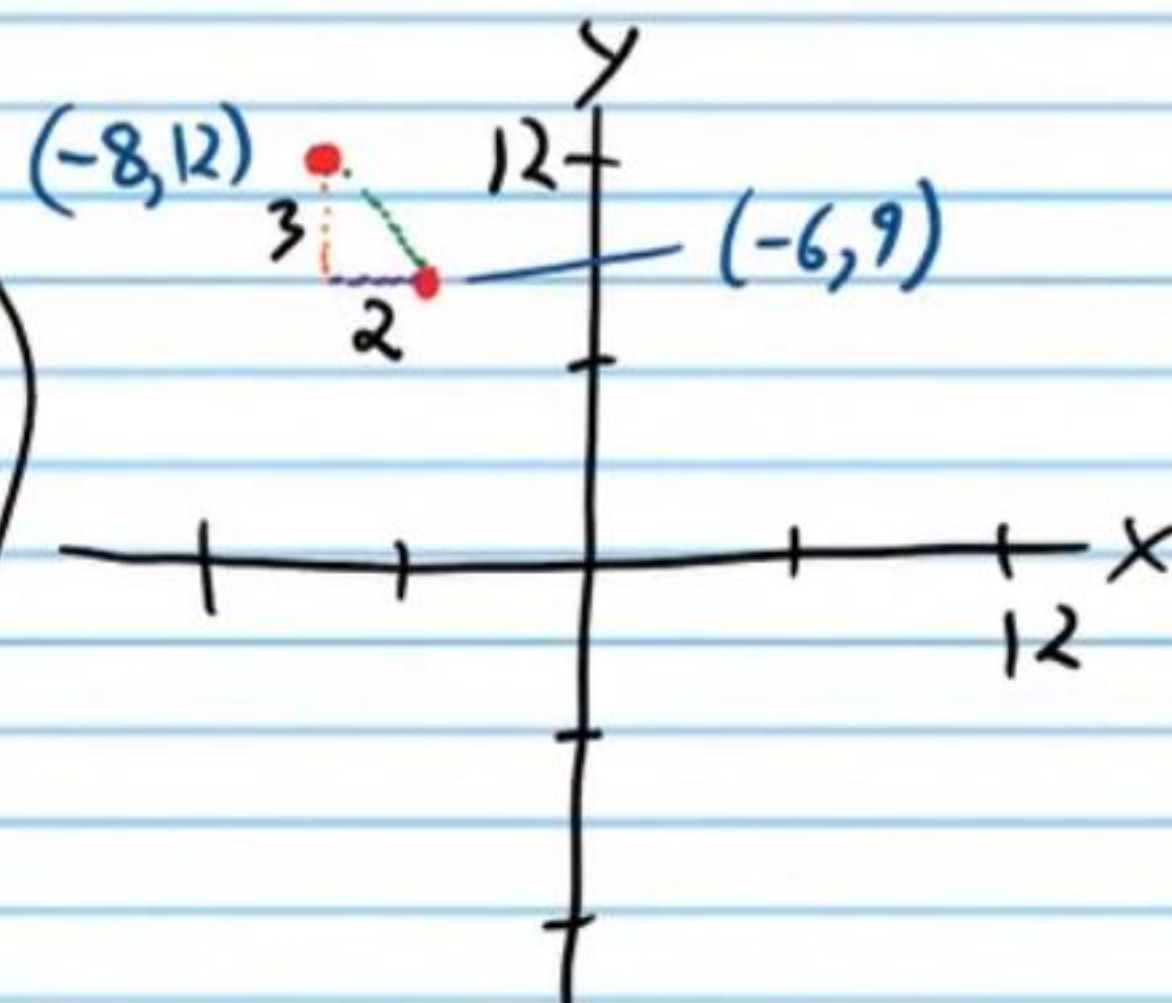
$$D = \sqrt{(-8 + 6)^2 + 3^2}$$

$$D = \sqrt{(-2)^2 + 3^2}$$

$$D = \sqrt{4 + 9}$$

$$\boxed{D = \sqrt{13}}$$

Graphical Method



$$a^2 + b^2 = c^2$$

$$2^2 + 3^2 = c^2$$

$$4 + 9 = c^2$$

$$13 = c^2$$

$$\sqrt{13} = \sqrt{c^2}$$

$$\sqrt{13} = c$$

$$\boxed{\sqrt{13} = D}$$

⑦  $(-3, -5)$  &  $(1, -3)$

Formula Method

$(x_1, y_1)$  &  $(x_2, y_2)$

$$D = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$D = \sqrt{(1 - (-3))^2 + (-3 - (-5))^2}$$

$$D = \sqrt{(1 + 3)^2 + (-3 + 5)^2}$$

$$D = \sqrt{4^2 + 2^2}$$

$$D = \sqrt{16 + 4}$$

$$D = \sqrt{20}$$

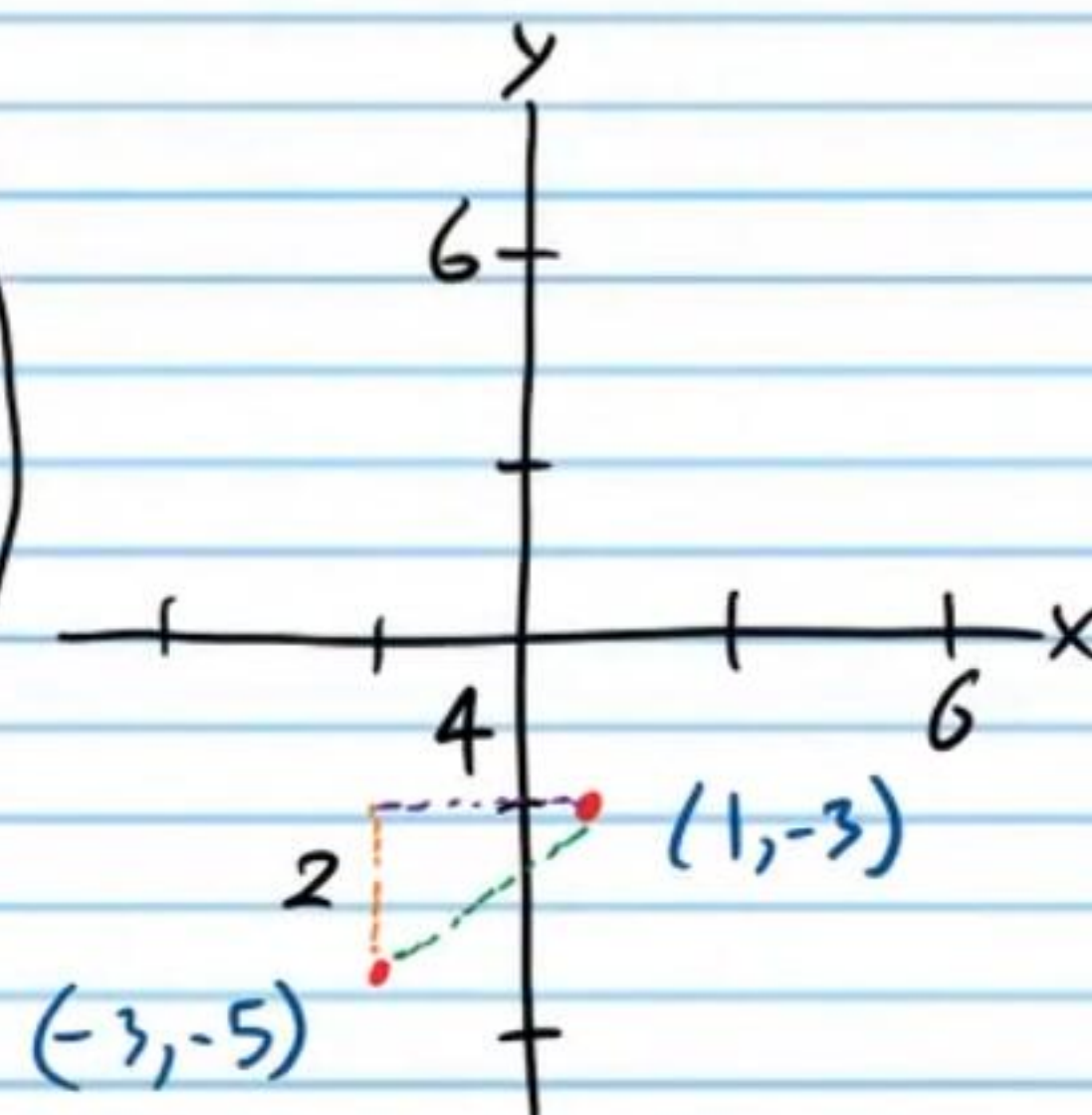
$$D = \sqrt{4 \cdot 5}$$

$$D = \sqrt{4} \cdot \sqrt{5}$$

$$D = 2 \cdot \sqrt{5}$$

$$\boxed{D = 2\sqrt{5}}$$

Graphical Method



$$a^2 + b^2 = c^2$$

$$4^2 + 2^2 = c^2$$

$$16 + 4 = c^2$$

$$20 = c^2$$

$$\sqrt{20} = \sqrt{c^2}$$

$$\sqrt{20} = c$$

$$\sqrt{4 \cdot 5} = c$$

$$\sqrt{4} \cdot \sqrt{5} = c$$

$$2 \cdot \sqrt{5} = c$$

$$\boxed{2\sqrt{5} = D}$$

⑧  $(-4, 0)$  &  $(4, 2)$

Formula Method

$(x_1, y_1)$  &  $(x_2, y_2)$

$$D = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$D = \sqrt{(4 - (-4))^2 + (2 - 0)^2}$$

$$D = \sqrt{(4 + 4)^2 + 2^2}$$

$$D = \sqrt{8^2 + 4}$$

$$D = \sqrt{64 + 4}$$

$$D = \sqrt{68}$$

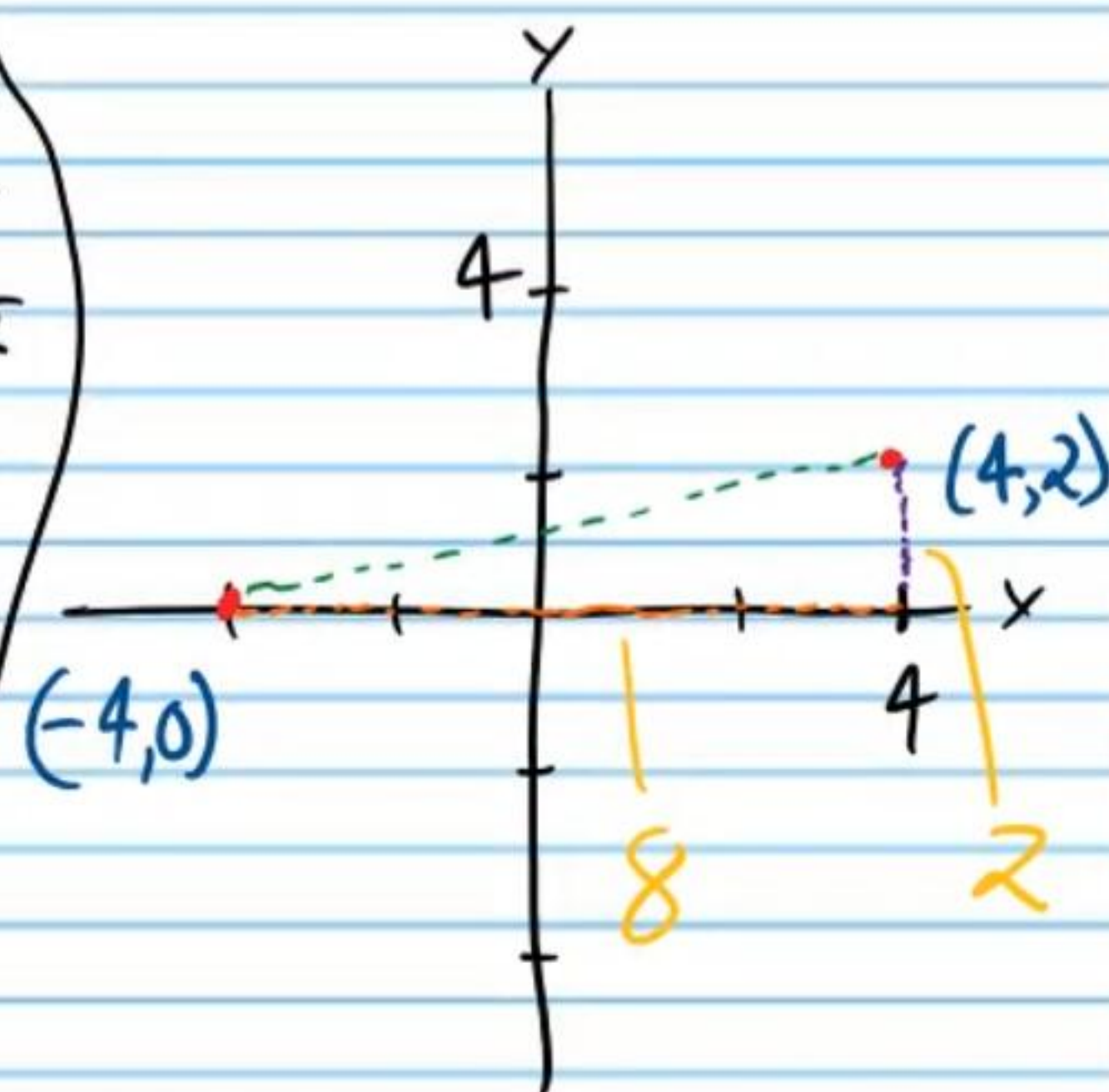
$$D = \sqrt{4 \cdot 17}$$

$$D = \sqrt{4} \cdot \sqrt{17}$$

$$D = 2 \cdot \sqrt{17}$$

$$\boxed{D = 2\sqrt{17}}$$

Graphical Method



$$a^2 + b^2 = c^2$$

$$8^2 + 2^2 = c^2$$

$$64 + 4 = c^2$$

$$68 = c^2$$

$$\sqrt{68} = \sqrt{c^2}$$

$$\sqrt{68} = c$$

$$\sqrt{4 \cdot 17} = c$$

$$\sqrt{4} \cdot \sqrt{17} = c$$

$$2 \cdot \sqrt{17} = c$$

$$\boxed{2\sqrt{17} = D}$$

⑨  $(4, 3)$  &  $(4, 6)$

Formula Method

$(x_1, y_1)$  &  $(x_2, y_2)$

$$D = \sqrt{(4 - 4)^2 + (6 - 3)^2}$$

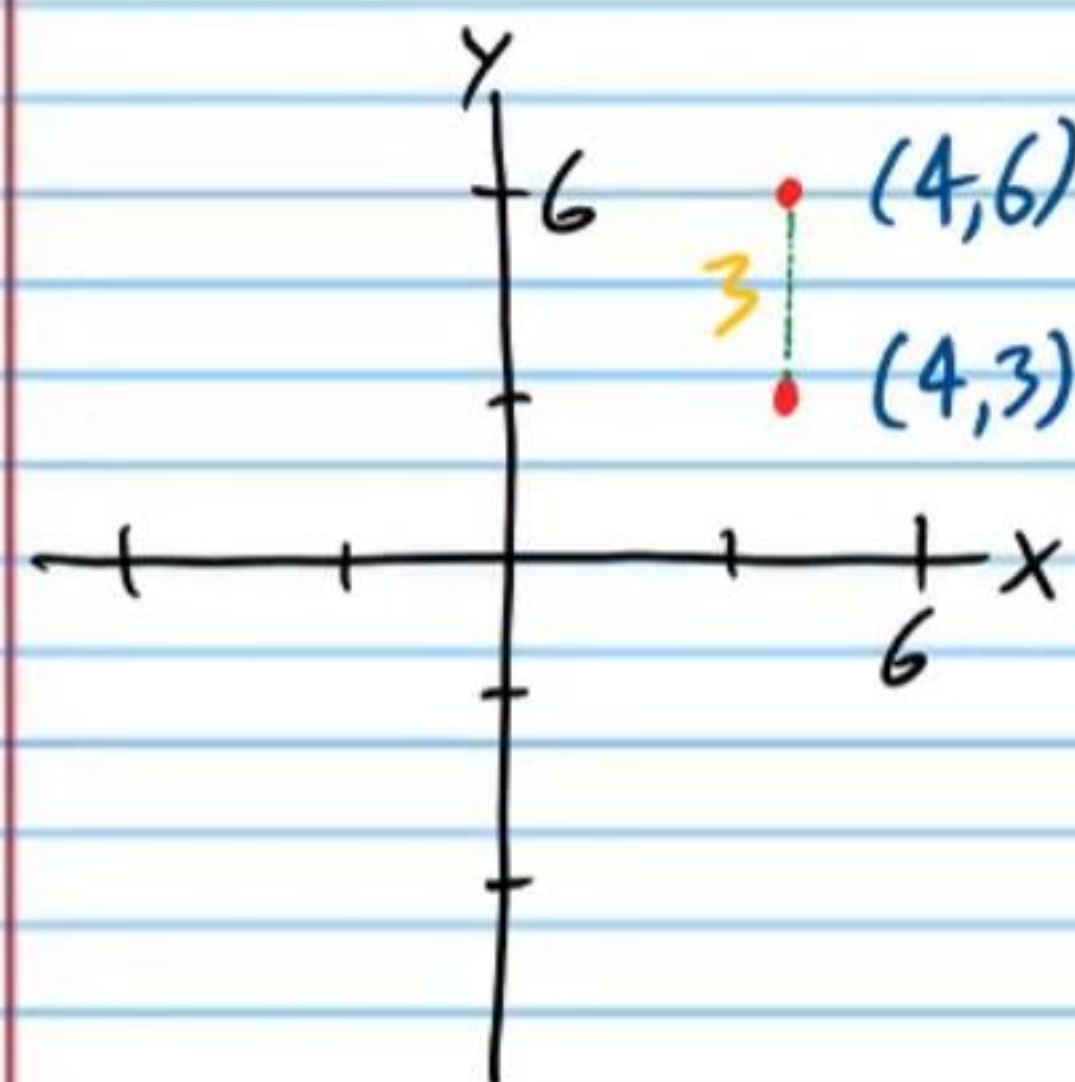
$$D = \sqrt{0^2 + 3^2}$$

$$D = \sqrt{0 + 9}$$

$$D = \sqrt{9}$$

$$\boxed{D = 3}$$

Graphical Method



$$D = 6 - 3 = \boxed{3}$$

⑩  $(-3, 5)$  &  $(4, 5)$

Formula Method

$(x_1, y_1)$  &  $(x_2, y_2)$

$$D = \sqrt{(4 - (-3))^2 + (5 - 5)^2}$$

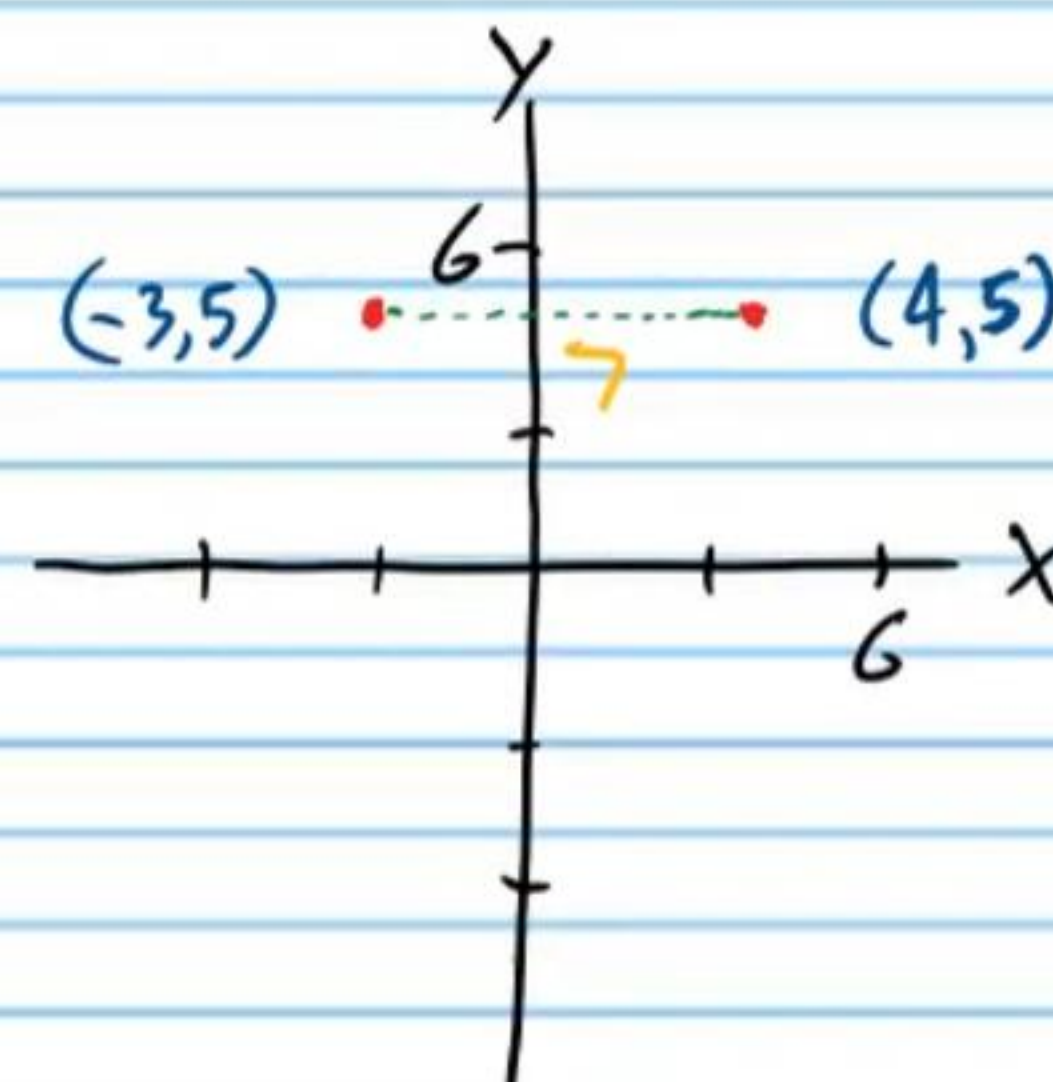
$$D = \sqrt{(4 + 3)^2 + 0^2}$$

$$D = \sqrt{7^2 + 0}$$

$$D = \sqrt{49 + 0}$$

$$D = \sqrt{49} = \boxed{7}$$

Graphical Method

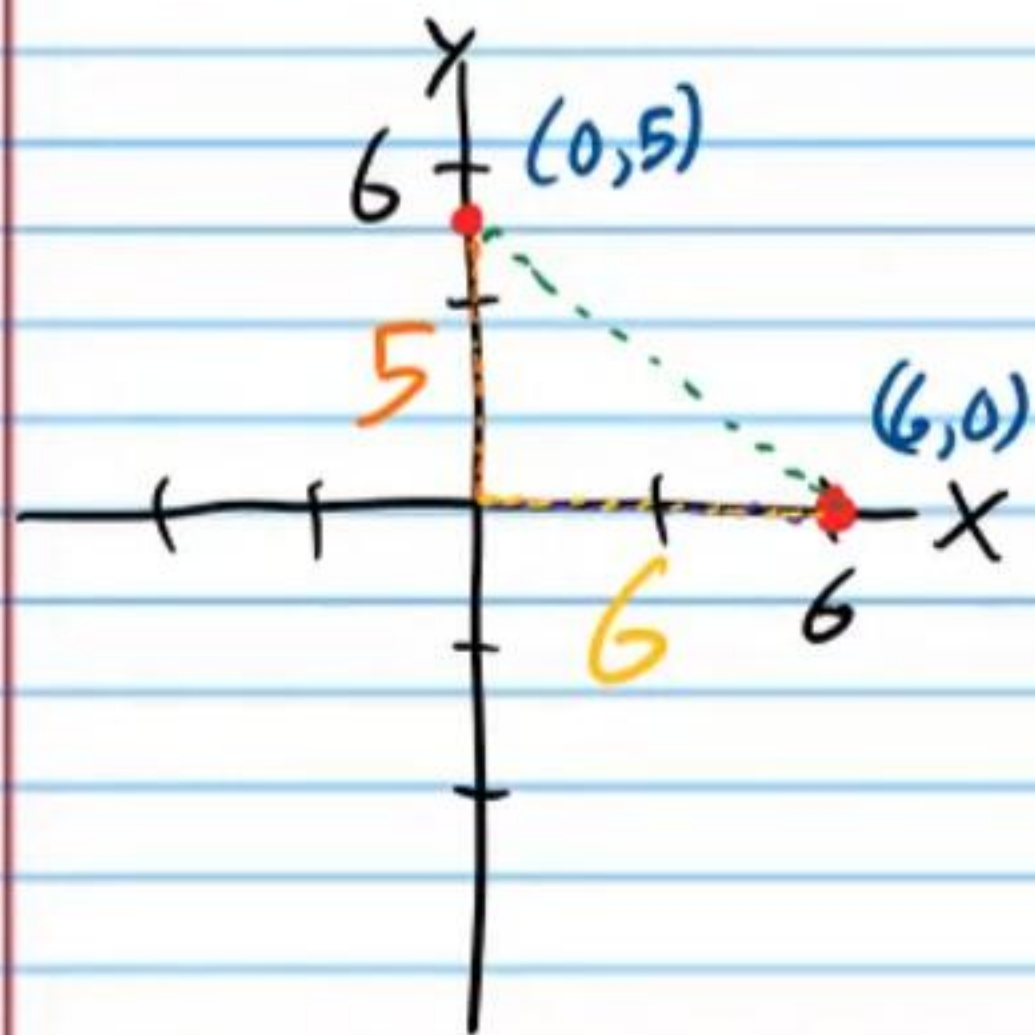


$$D = -3 - 4 = -7$$

$$\boxed{D = 7}$$

⑪ (0,5) & (6,0)

Graphical Method



$$a^2 + b^2 = c^2$$

$$5^2 + 6^2 = c^2$$

$$25 + 36 = c^2$$

$$61 = c^2$$

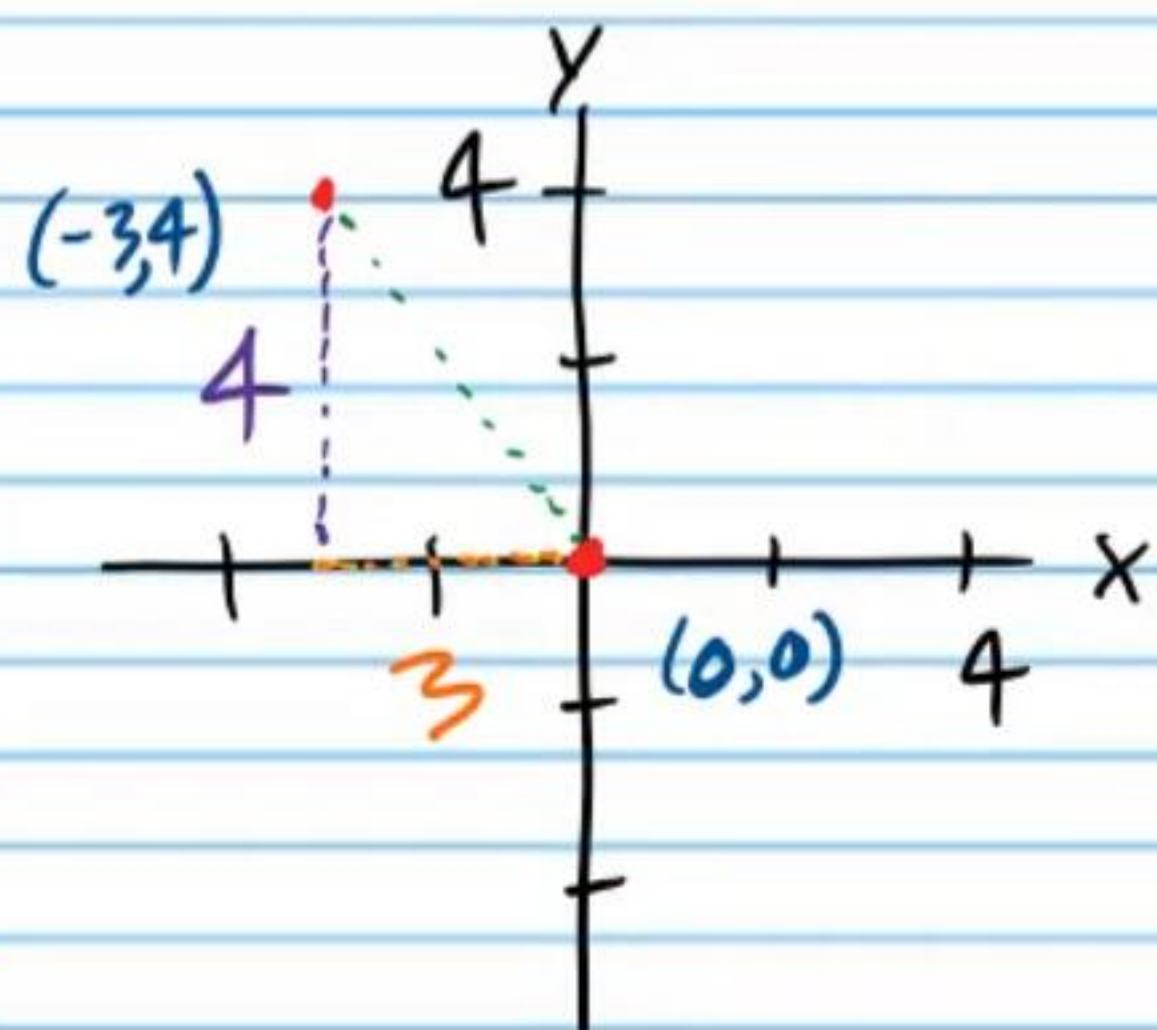
$$\sqrt{61} = \sqrt{c^2}$$

$$\sqrt{61} = c$$

$$\boxed{\sqrt{61} = D}$$

⑫ (-3,4) & (0,0)

Graphical Method



$$a^2 + b^2 = c^2$$

$$4^2 + 3^2 = c^2$$

$$16 + 9 = c^2$$

$$25 = c^2$$

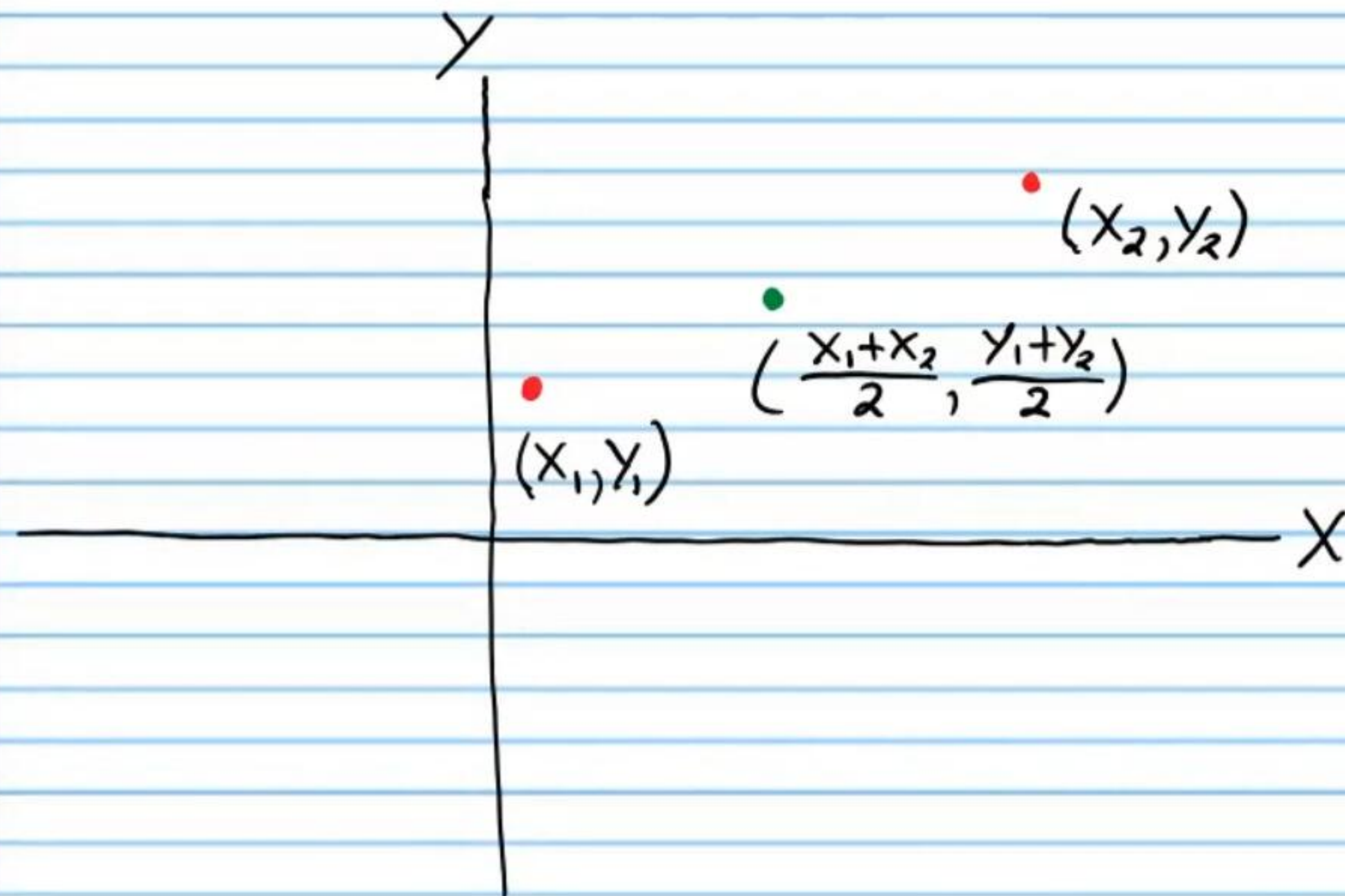
$$\sqrt{25} = \sqrt{c^2}$$

$$5 = c$$

$$\boxed{5 = D}$$

## The Midpoint

The midpoint between points  $(x_1, y_1)$  and  $(x_2, y_2)$  is  $\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$ .



Find the midpoint between the following coordinates.

⑬  $(3, -2)$  &  $(6, 1)$

$(x_1, y_1)$  &  $(x_2, y_2)$

$M\left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right)$

$M\left(\frac{3+6}{2}, \frac{-2+1}{2}\right)$

$M\left(\frac{9}{2}, -\frac{1}{2}\right)$

⑭  $(-8, 6)$  &  $(4, -3)$

$(x_1, y_1)$  &  $(x_2, y_2)$

$M\left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right)$

$M\left(\frac{-8+4}{2}, \frac{6+(-3)}{2}\right)$

$M\left(-\frac{4}{2}, \frac{3}{2}\right)$

$M\left(-2, \frac{3}{2}\right)$

⑮  $(-2, -7)$  &  $(-1, -1)$

$(x_1, y_1)$  &  $(x_2, y_2)$

$M\left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right)$

$M\left(\frac{-2+(-1)}{2}, \frac{-7+(-1)}{2}\right)$

$M\left(-\frac{3}{2}, -\frac{8}{2}\right)$

$M\left(-\frac{3}{2}, -4\right)$

⑯  $(5, -6)$  &  $(0, -4)$

$(x_1, y_1)$  &  $(x_2, y_2)$

$M\left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right)$

$M\left(\frac{5+0}{2}, \frac{-6+(-4)}{2}\right)$

$M\left(\frac{5}{2}, -\frac{10}{2}\right)$

$M\left(\frac{5}{2}, -5\right)$

⑰  $(3, 7)$  &  $(1, 4)$

$(x_1, y_1)$  &  $(x_2, y_2)$

$M\left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right)$

$M\left(\frac{3+1}{2}, \frac{7+4}{2}\right)$

$M\left(\frac{4}{2}, \frac{11}{2}\right)$

$M\left(2, \frac{11}{2}\right)$

⑱  $(-7, 4)$  &  $(5, -3)$

$(x_1, y_1)$  &  $(x_2, y_2)$

$M\left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right)$

$M\left(\frac{-7+5}{2}, \frac{4+(-3)}{2}\right)$

$M\left(-\frac{2}{2}, \frac{1}{2}\right)$

$M\left(-1, \frac{1}{2}\right)$

⑲  $(-2, -2)$  &  $(-5, -3)$

$(x_1, y_1)$  &  $(x_2, y_2)$

$M\left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right)$

$M\left(\frac{-2+(-5)}{2}, \frac{-2+(-3)}{2}\right)$

$M\left(-\frac{7}{2}, -\frac{5}{2}\right)$

⑳  $(0, 4)$  &  $(8, 0)$

$(x_1, y_1)$  &  $(x_2, y_2)$

$M\left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right)$

$M\left(\frac{0+8}{2}, \frac{4+0}{2}\right)$

$M\left(\frac{8}{2}, \frac{4}{2}\right)$

$M(4, 2)$

② Find the distance between Lana's house and her favorite restaurant shown on the grid below. The units on the grid represent miles.

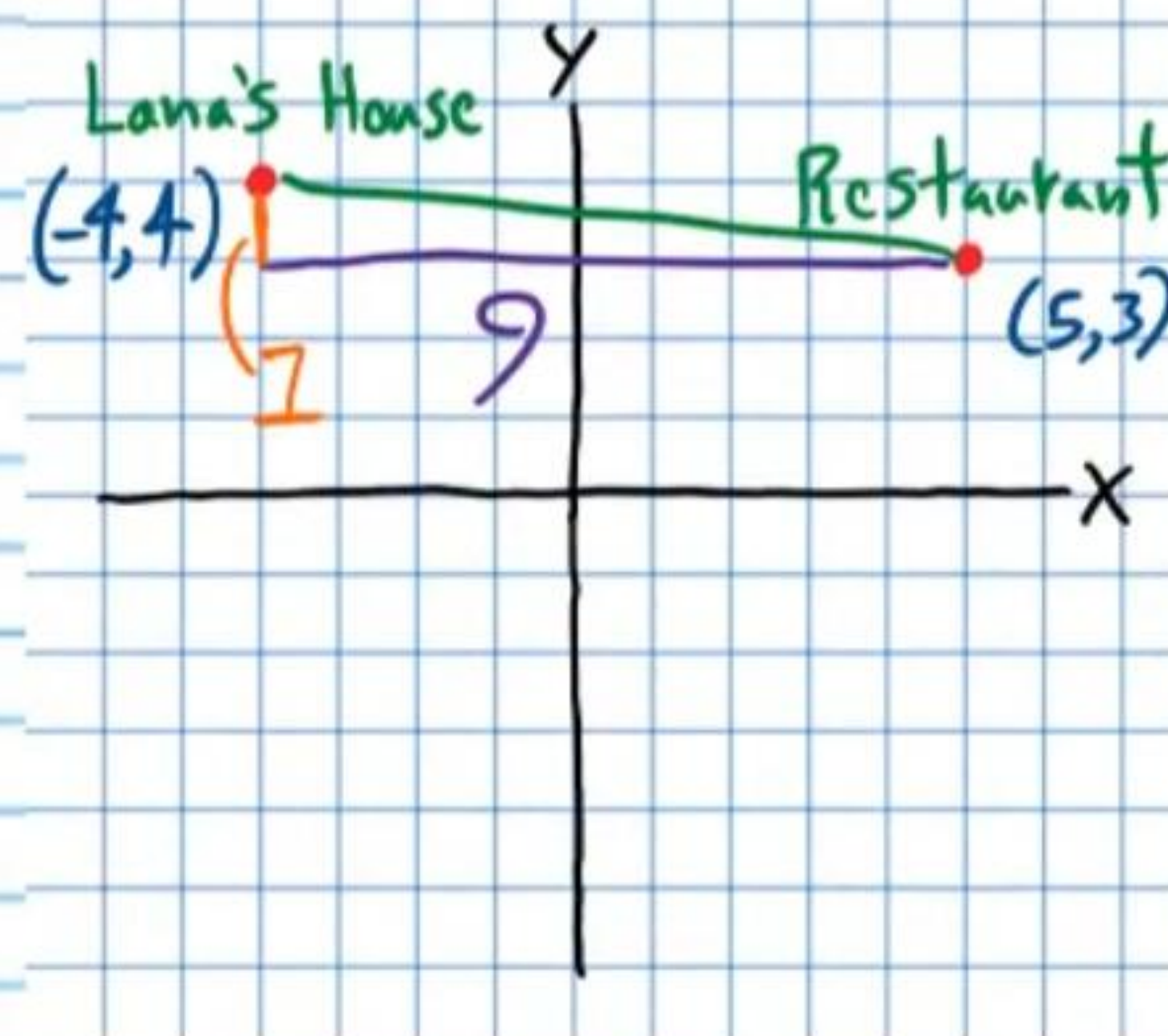
$$D = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$D = \sqrt{(5 - (-4))^2 + (3 - 4)^2}$$

$$D = \sqrt{9^2 + (-1)^2}$$

$$D = \sqrt{81 + 1}$$

$$D = \sqrt{82} \text{ mi}$$



$$a^2 + b^2 = c^2$$

$$1^2 + 9^2 = c^2$$

$$1 + 81 = c^2$$

$$82 = c^2$$

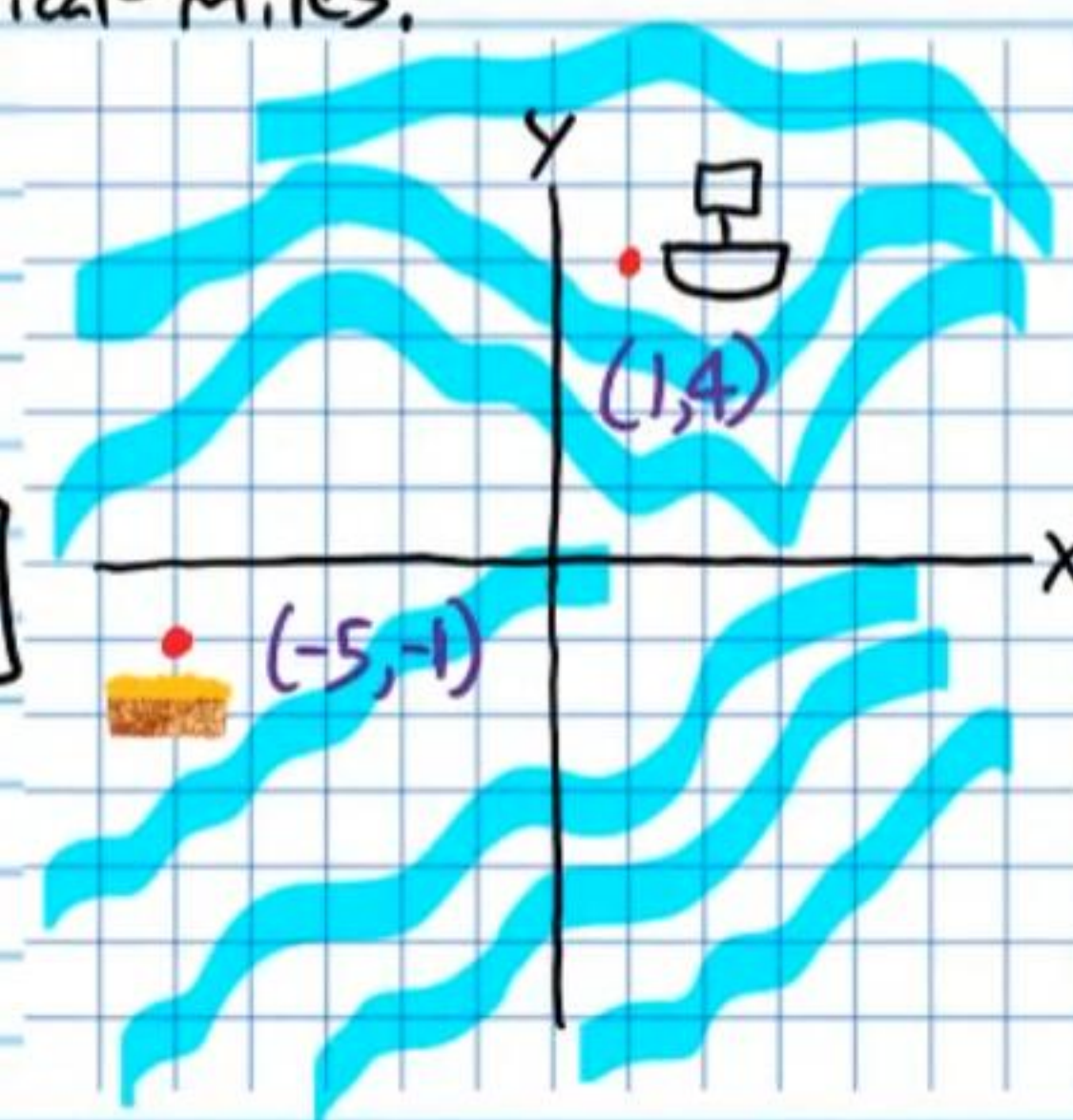
$$\sqrt{82} \text{ mi} = D$$

② Find the distance between the pirate and his treasure shown on the grid below. The units on the grid represent nautical miles.

$$D = \sqrt{(4 - (-1))^2 + (1 - (-5))^2}$$

$$D = \sqrt{25 + 36}$$

$$D = \sqrt{61} \text{ nautical miles}$$



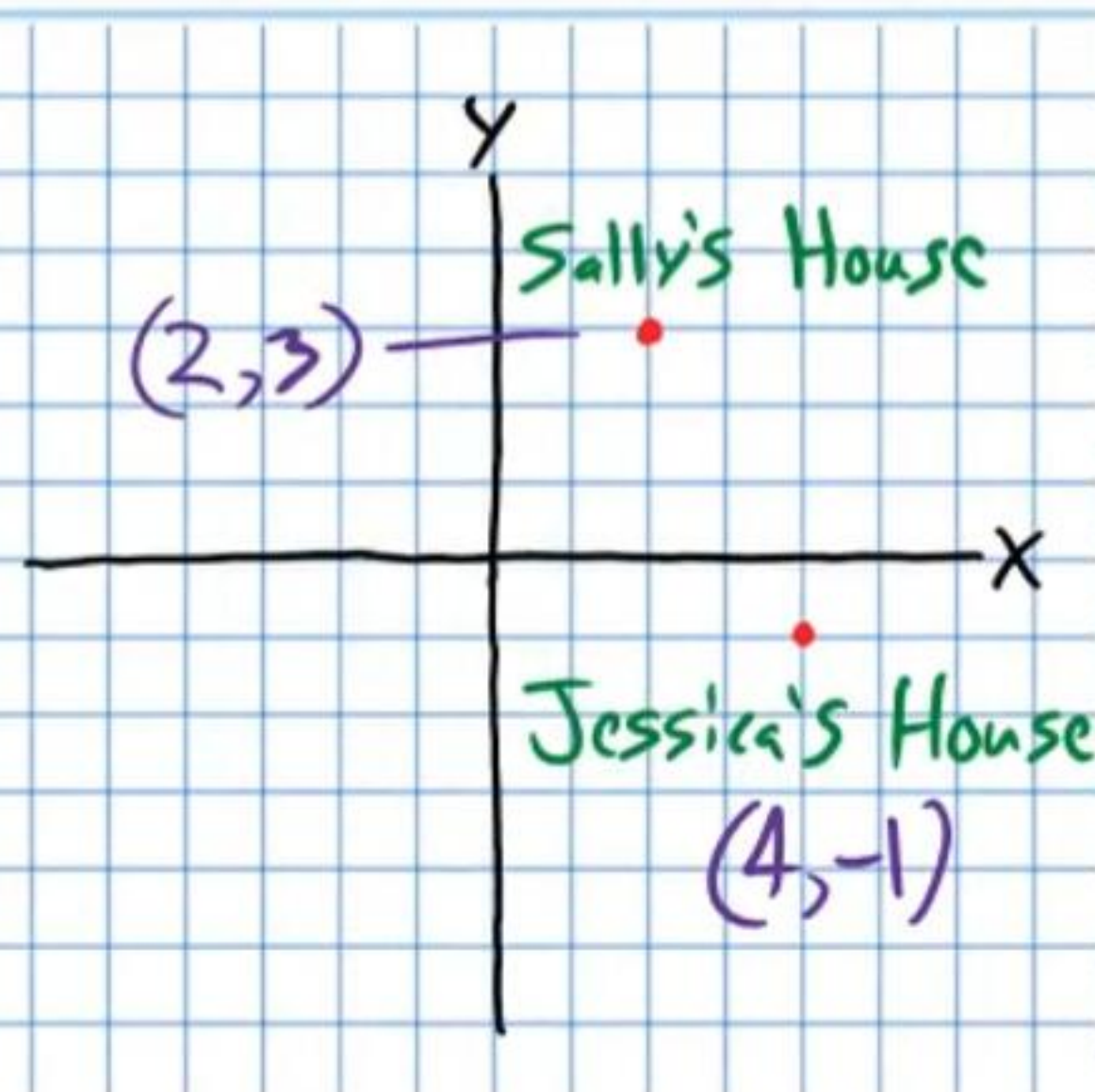
②③ Sally and Jessica want to meet halfway between their homes to have lunch (locations shown below). Find the coordinates of their meeting place.

$$M\left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right)$$

$$M\left(\frac{2+4}{2}, \frac{3+(-1)}{2}\right)$$

$$M\left(\frac{6}{2}, \frac{2}{2}\right)$$

$$\boxed{M(3, 1)}$$

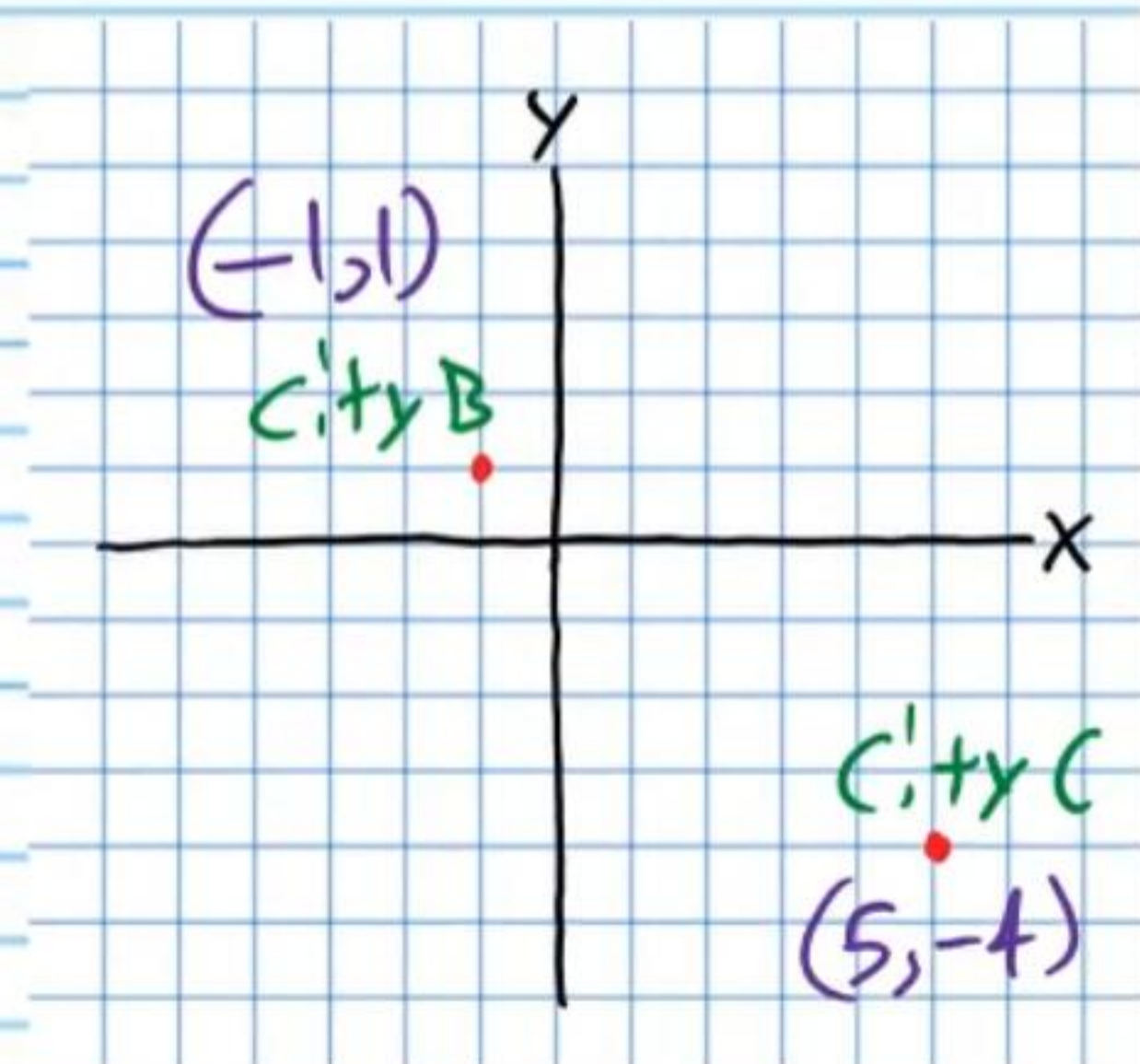


②④ City A is halfway between City B and City C (locations shown below). Find the coordinates of City A.

$$M\left(\frac{-1+5}{2}, \frac{1+(-4)}{2}\right)$$

$$M\left(\frac{4}{2}, \frac{-3}{2}\right)$$

$$\boxed{M(2, -\frac{3}{2})}$$



②5 Find the distance between two ant hills shown below on the grid. Each unit on the grid represents five feet.

$$D = \sqrt{(-3 - (-1))^2 + (-1 - 1)^2}$$

$$D = \sqrt{4 + 4}$$

$$D = \sqrt{8}$$

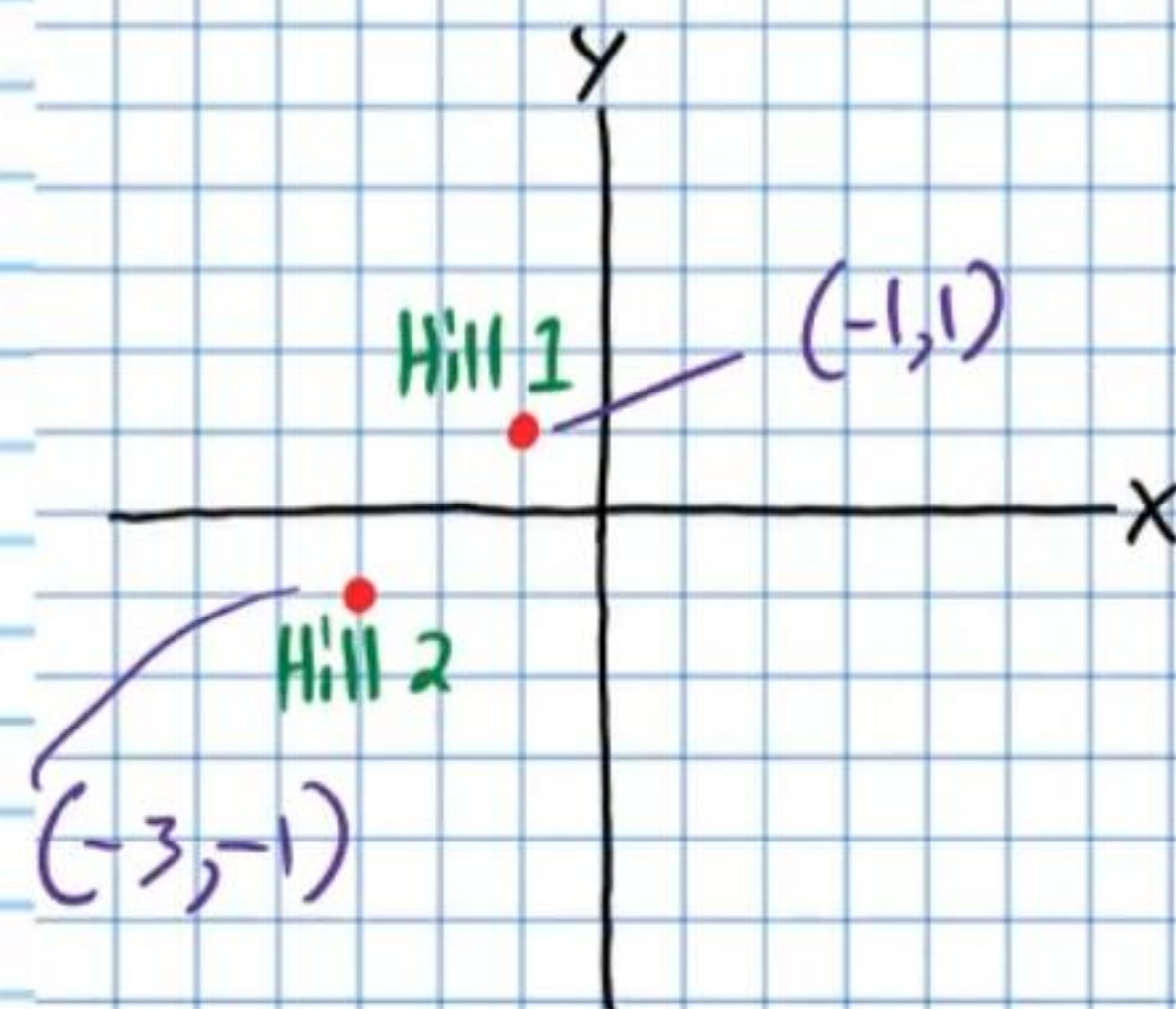
$$D = \sqrt{4 \cdot 2}$$

$$D = \sqrt{4} \cdot \sqrt{2}$$

$$D = 2\sqrt{2}$$

↓

$$D_A = 5 \cdot 2\sqrt{2} = \boxed{10\sqrt{2} \text{ ft}}$$



②6 Find the distance between two gas stations shown on the grid below. Each unit on the grid represents three kilometers.

$$D = \sqrt{(-1 - 0)^2 + (-3 - 4)^2}$$

$$D = \sqrt{1 + 49}$$

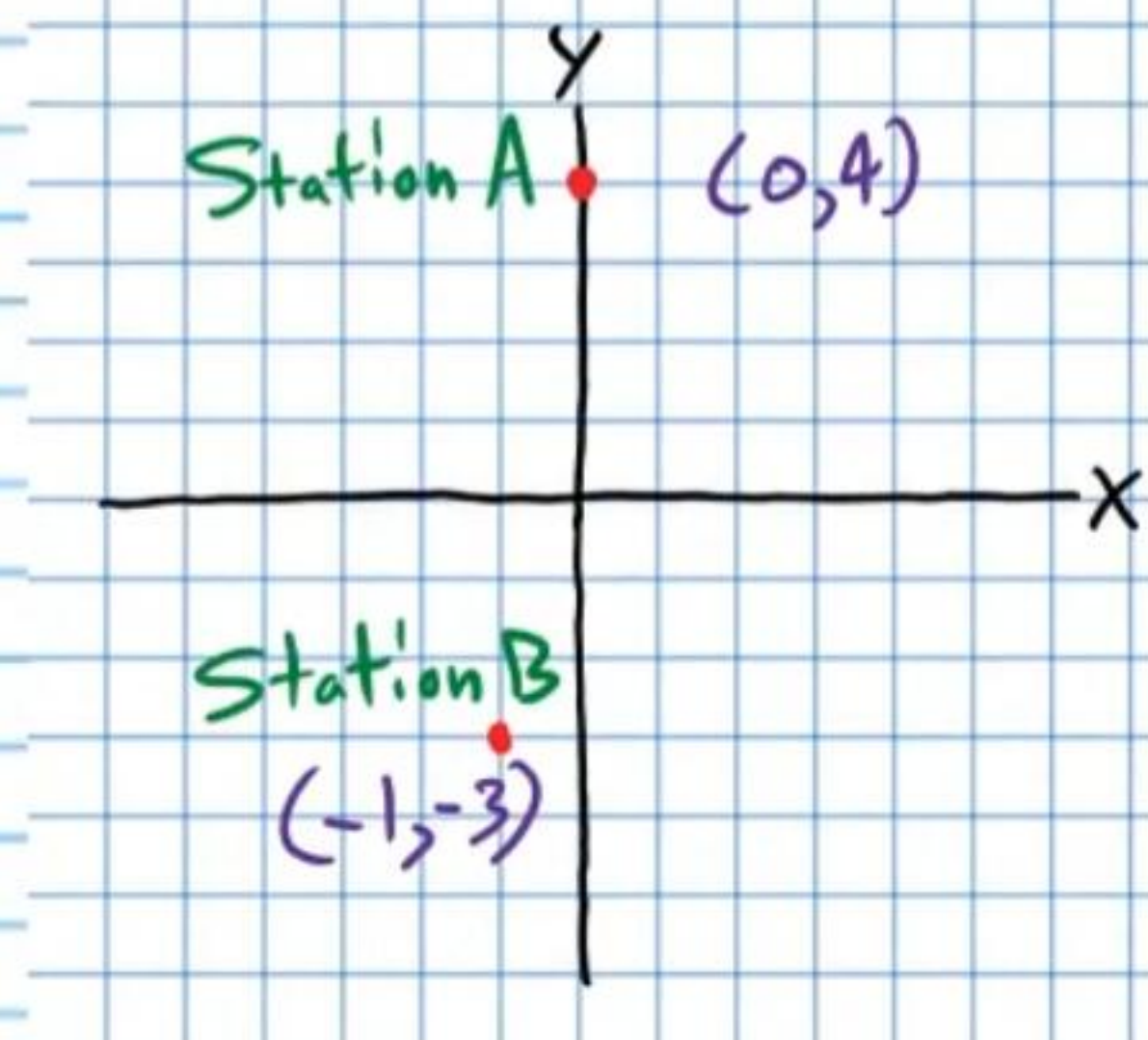
$$D = \sqrt{50}$$

$$D = \sqrt{25 \cdot 2}$$

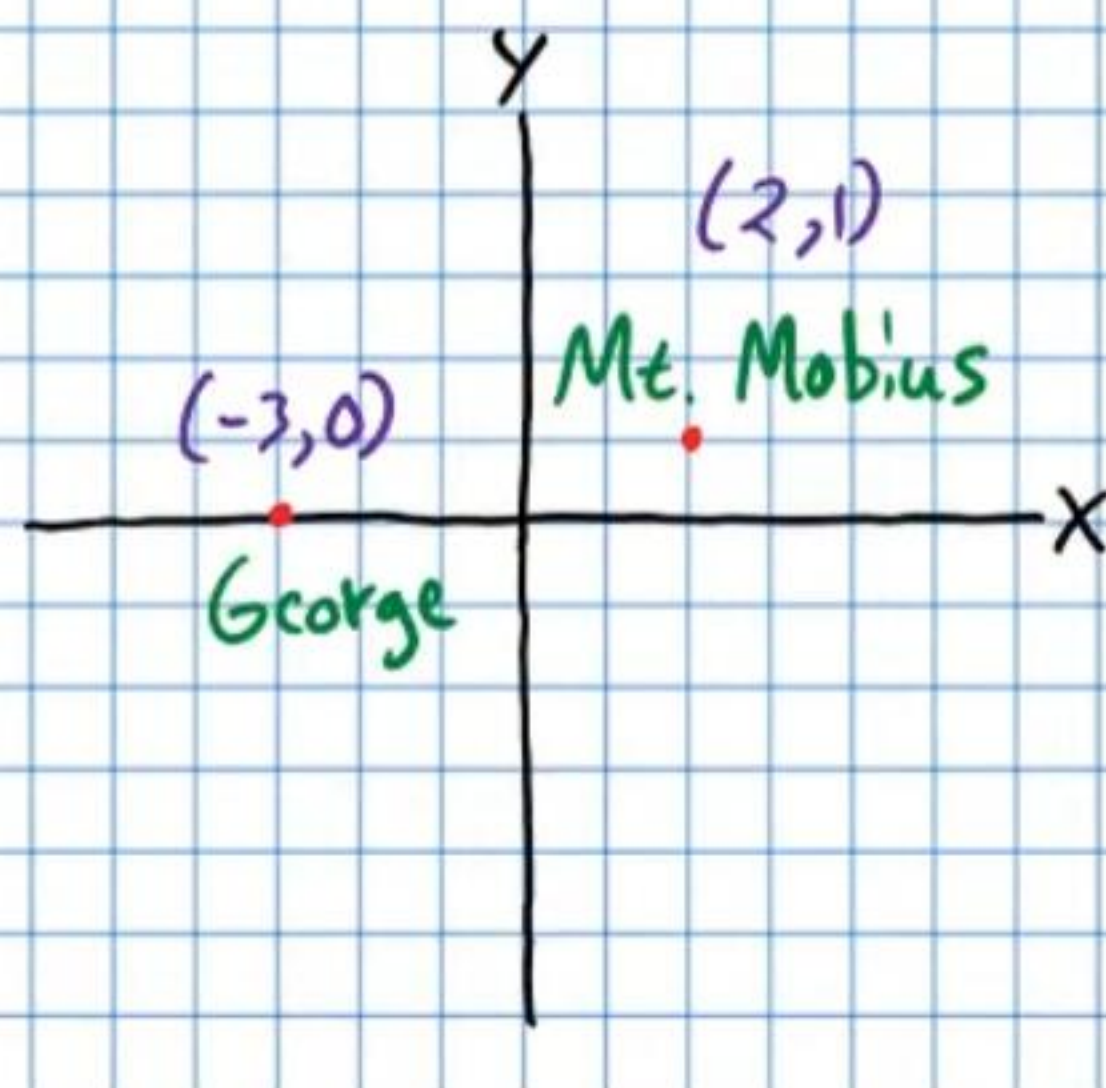
$$D = \sqrt{25} \cdot \sqrt{2}$$

$$D = 5\sqrt{2}$$

$$D_A = 3 \cdot 5\sqrt{2} = \boxed{15\sqrt{2} \text{ km}}$$



(27) George wants to hike to Mount Mobius, but he will need two days to make the trip. If he wants to hike half the distance in the first day, how far will he hike on the first day (each unit on the grid represents four miles)?



$$D = \sqrt{(-3-2)^2 + (0-1)^2}$$

$$D = \sqrt{25 + 1}$$

$$D = \sqrt{26}$$

$$D_A = \frac{4 \cdot \sqrt{26}}{2} \text{ mi}$$

$$D_A = 2 \cdot \sqrt{26} \text{ mi}$$

$$D_A = 2\sqrt{26} \text{ mi}$$

## Challenge Problems

(28) If (2, 3) is the midpoint between points A and B, and the coordinates of B are (1, 1), what are the coordinates of point A?

$$M\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$$

$$\frac{x_1 + x_2}{2} = x_m \quad \frac{y_1 + y_2}{2} = y_m$$

$$\frac{1 + x_2}{2} = 2 \quad \frac{1 + y_2}{2} = 3$$

$$2 \cdot \frac{1 + x_2}{2} = 2 \cdot 2 \quad 1 + y_2 = 6$$

$$1 + x_2 = 4 \quad y_2 = 5$$

$$x_2 = 3$$

$$A(3, 5)$$

(29) If (7, -2) is the midpoint between points C and D, and the coordinates of C are (-3, -4), what are the coordinates of point D?

$$\frac{x_1 + x_2}{2} = x_m \quad \frac{y_1 + y_2}{2} = y_m$$

$$\frac{-3 + x_2}{2} = 7 \quad \frac{-4 + y_2}{2} = -2$$

$$-3 + x_2 = 14 \quad -4 + y_2 = -4$$

$$x_2 = 17 \quad y_2 = 0$$

$$D(17, 0)$$

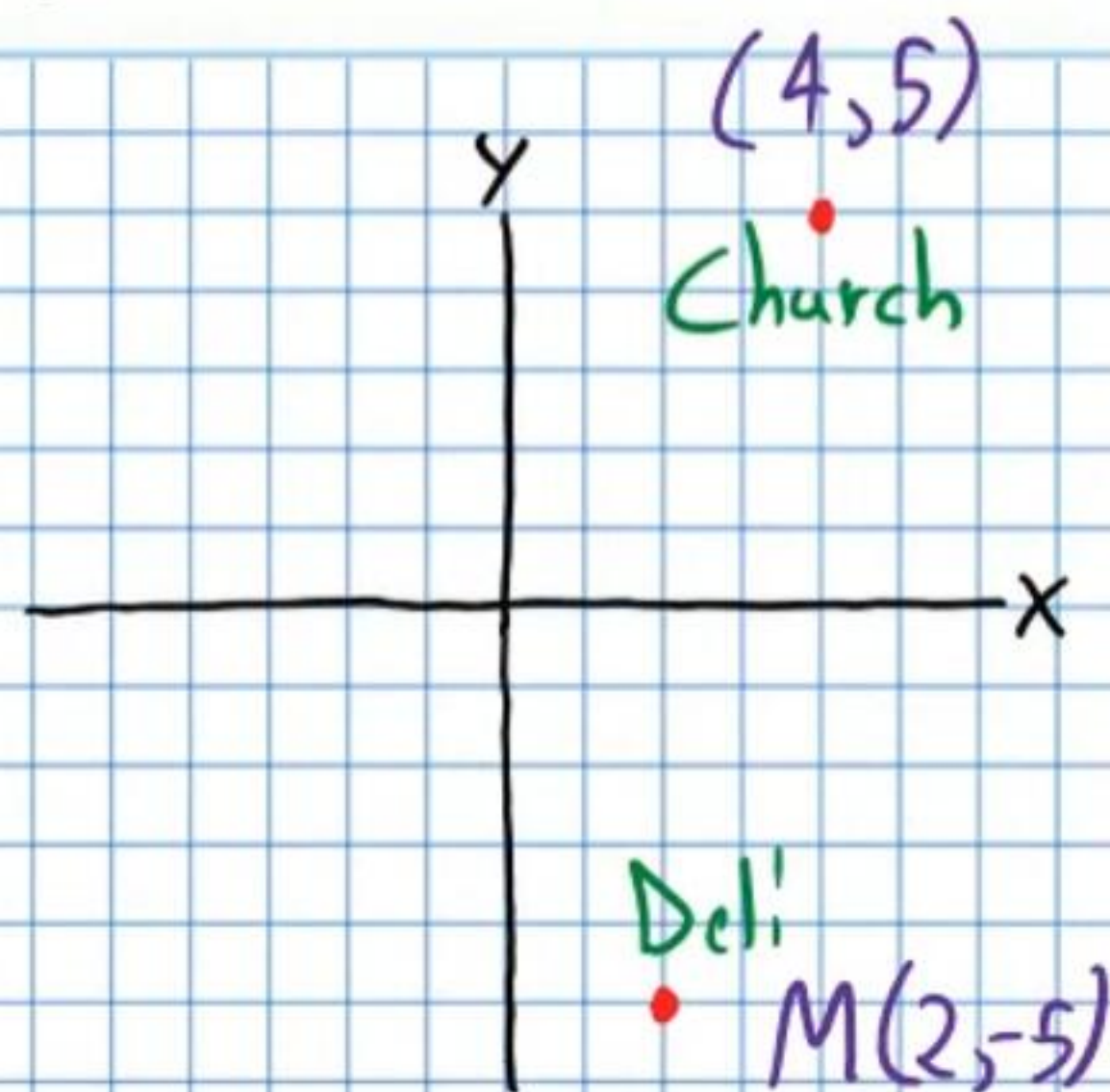
③⑩ If the deli is halfway between the museum and the church, find the coordinates of the museum.

$$\frac{x_1 + x_2}{2} = x_m$$

$$\frac{4 + x_2}{2} = 2$$

$$4 + x_2 = 4$$

$$x_2 = 0$$



$$\frac{y_1 + y_2}{2} = y_m$$

$$\frac{5 + y_2}{2} = -5$$

$$5 + y_2 = -10$$

$$y_2 = -15$$

**(0, -15)**

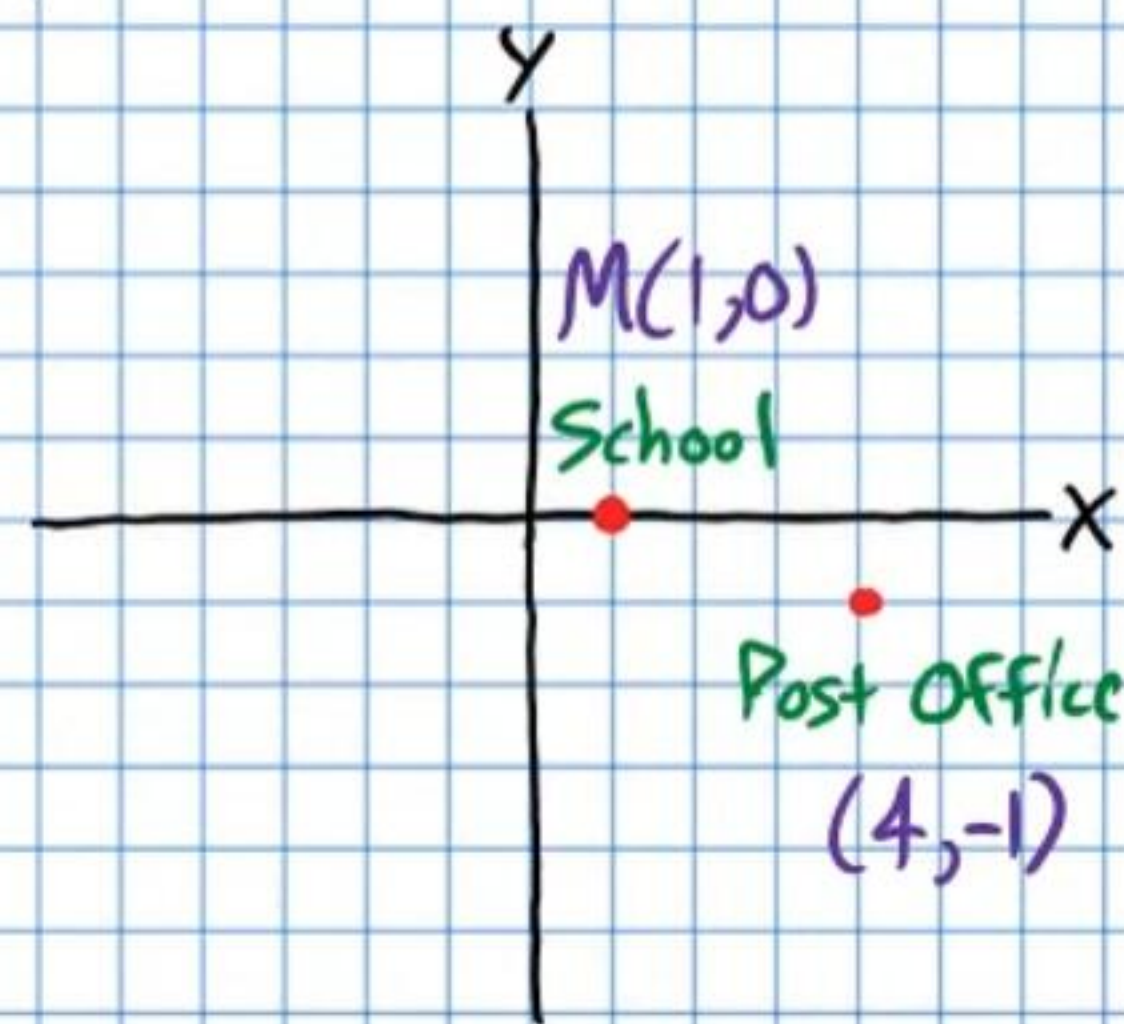
③⑪ If the school is halfway between the factory and the post office, what are the coordinates of the factory?

$$\frac{x_1 + x_2}{2} = x_m$$

$$\frac{4 + x_2}{2} = 1$$

$$4 + x_2 = 2$$

$$x_2 = -2$$



$$\frac{y_1 + y_2}{2} = y_m$$

$$\frac{-1 + y_2}{2} = 0$$

$$-1 + y_2 = 0$$

$$y_2 = 1$$

**(-2, 1)**

32) If the distance between  $(-4, 1)$  and  $(2, y)$  is  $\sqrt{37}$ , Find the possible values of  $y$ .

$$(-4, 1) \& (2, y)$$

$$(x_1, y_1) \& (x_2, y_2)$$

$$D = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$\sqrt{37} = \sqrt{(2 - (-4))^2 + (y - 1)^2}$$

$$(\sqrt{37})^2 = (\sqrt{(2 - (-4))^2 + (y - 1)^2})^2$$

$$37 = (2 - (-4))^2 + (y - 1)^2$$

$$37 = (2 + 4)^2 + (y - 1)^2$$

$$37 = 6^2 + (y - 1)^2$$

$$37 = 36 + (y - 1)^2$$

$$1 = (y - 1)^2$$

$$\pm\sqrt{1} = \sqrt{(y - 1)^2}$$

$$\pm 1 = y - 1$$

$$-1 = y - 1$$

$$\boxed{0 = y}$$

$$1 = y - 1$$

$$\boxed{2 = y}$$

33) If the distance between  $(x, 1)$  and  $(2, 3)$  is  $2\sqrt{17}$ , Find the possible values of  $x$ .

$$(x, 1) \& (2, 3)$$

$$(x_1, y_1) \& (x_2, y_2)$$

$$D = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$2\sqrt{17} = \sqrt{(2 - x)^2 + (3 - 1)^2}$$

$$(2\sqrt{17})^2 = (\sqrt{(2 - x)^2 + (3 - 1)^2})^2$$

$$4 \cdot 17 = (2 - x)^2 + (3 - 1)^2$$

$$68 = (2 - x)^2 + 2^2$$

$$68 = (2 - x)^2 + 4$$

$$64 = (2 - x)^2$$

$$\pm\sqrt{64} = \sqrt{(2 - x)^2}$$

$$\pm 8 = 2 - x$$

$$-8 = 2 - x$$

$$-10 = -x$$

$$\boxed{10 = x}$$

$$8 = 2 - x$$

$$6 = -x$$

$$\boxed{-6 = x}$$

34) If the distance between  $(-5, 3)$  and  $(x, -1)$  is  $\sqrt{65}$ , Find the possible values of  $x$ .

$$(-5, 3) \& (x, -1)$$

$$(x_1, y_1) \& (x_2, y_2)$$

$$D = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$\sqrt{65} = \sqrt{(x - (-5))^2 + (-1 - 3)^2}$$

$$(\sqrt{65})^2 = (\sqrt{(x - (-5))^2 + (-1 - 3)^2})^2$$

$$65 = (x - (-5))^2 + (-1 - 3)^2$$

$$65 = (x + 5)^2 + (-4)^2$$

$$65 = (x + 5)^2 + 16$$

$$49 = (x + 5)^2$$

$$\pm\sqrt{49} = \sqrt{(x + 5)^2}$$

$$\pm 7 = x + 5$$

$$-7 = x + 5$$

$$\boxed{-12 = x}$$

$$7 = x + 5$$

$$\boxed{2 = x}$$

35) If the distance between  $(5, y)$  and  $(0, 2)$  is  $5\sqrt{5}$ , Find the possible values of  $y$ .

$$(5, y) \& (0, 2)$$

$$(x_1, y_1) \& (x_2, y_2)$$

$$D = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$5\sqrt{5} = \sqrt{(0 - 5)^2 + (2 - y)^2}$$

$$(5\sqrt{5})^2 = (\sqrt{(0 - 5)^2 + (2 - y)^2})^2$$

$$25 \cdot 5 = (0 - 5)^2 + (2 - y)^2$$

$$125 = (-5)^2 + (2 - y)^2$$

$$125 = 25 + (2 - y)^2$$

$$100 = (2 - y)^2$$

$$\pm\sqrt{100} = \sqrt{(2 - y)^2}$$

$$\pm 10 = 2 - y$$

$$-10 = 2 - y$$

$$-12 = -y$$

$$\boxed{12 = y}$$

$$10 = 2 - y$$

$$8 = -y$$

$$\boxed{-8 = y}$$